



BÁO CÁO ĐỒ ÁN MÔN HỌC

ĐỀ TÀI: SHORT-TERM BITCOIN RETURN PREDICTION BASED ON MULTI-SOURCE DATA

Môn học: Nhập môn khoa học dữ liệu

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ABSTRACT

Short-term bitcoin price movement forecasting is a challenging problem due to the market's 24/7 operation, high volatility, and inherently low signal-to-noise ratio. This report presents the development of a bitcoin return prediction system on the 4-hour (H4) timeframe by integrating data from seven heterogeneous sources, including market data (OHLCV), market capitalization, network activity, mining indicators, on-chain metrics, news data processed using CryptoBERT, and market sentiment indicators (Fear & Greed Index).

The study applies a rigorous feature extraction and feature engineering pipeline to construct 31 predictive variables while ensuring stationarity and preventing future information leakage. By evaluating ten different machine learning models on historical data spanning from 2018 to 2025, the empirical results demonstrate that regularized linear models consistently outperform more complex nonlinear models in a noisy financial environment.

Specifically, the Lasso regression model achieves the best performance on the test set (out-of-sample data), with a Spearman information coefficient (IC) of 0.269 and a directional accuracy of 57.86%. In a realistic trading simulation, the proposed system delivers a total return of +528% (corresponding to a simulated PnL of +5.28 billion USD), significantly outperforming the baseline strategy while maintaining effective risk control, with a maximum drawdown of only -9%. These findings not only provide a practical and robust framework for financial forecasting but also highlight the value of combining alternative data sources and machine learning techniques in quantitative trading.

Keywords: bitcoin forecasting, machine learning, multi-source data, lasso regression, on-chain metrics, sentiment analysis, quantitative trading.

¹ Github repository: [phatv1/crypto-price-forecaster](https://github.com/phatv1/crypto-price-forecaster)

CONTRIBUTION

Vũ Hoàng Phát	Responsible for collecting, cleaning, and performing exploratory data analysis (EDA) on OHLCV and news data; engineering features from price and news data; training, evaluating, and selecting machine learning models.
Phạm Nguyễn Quốc Huy	Responsible for collecting and processing market capitalization data; analyzing network activity; handling mining and security-related metrics.
Trần Phi Hùng	Responsible for collecting and processing advanced on-chain data; constructing and integrating on-chain indicators; processing and integrating market sentiment indices.

CHAPTER I. OVERVIEW

1. Introduction

1.1. Research background and motivation

The cryptocurrency market has experienced rapid growth over the past decade, with global market capitalization surpassing USD 1 trillion in 2021. Bitcoin, as the market's leading asset, has consistently attracted significant attention from both retail and institutional investors. However, short-term bitcoin price forecasting poses substantial challenges due to the unique characteristics of this market.

Unlike traditional financial markets, the cryptocurrency market exhibits extremely high volatility (bitcoin prices can fluctuate by 10–20% within 24 hours), operates continuously on a 24/7 basis without fixed trading sessions, and displays nonlinear and non-stationary dynamics. In addition, price movements are influenced by multiple heterogeneous data sources, including market data, on-chain activity, media news, and investor sentiment.

These characteristics limit the effectiveness of traditional technical analysis and statistical methods. In particular, many classical models rely on stationarity assumptions that are severely violated in cryptocurrency time series. Moreover, conventional linear models often fail to capture complex nonlinear patterns, while the effects of market sentiment and information flows are typically insufficiently incorporated within these analytical frameworks.

In this context, machine learning and multi-source data analysis have emerged as promising approaches for addressing short-term forecasting problems. Integrating market data, on-chain data, news information, and sentiment indicators is expected to produce more informative feature sets, thereby significantly improving the predictive performance of short-term cryptocurrency price movements.

1.2. Research problem

This study focuses on addressing the core research question: how can a short-term bitcoin return prediction system on a 4-hour timeframe be constructed with high accuracy and practical applicability? To achieve this objective, the following specific research questions are investigated:

- (1) How can data from multiple sources (market, on-chain, news, sentiment) be effectively integrated to construct predictive feature sets?
- (2) Do the engineered features adequately reflect market dynamics and satisfy the stationarity requirements necessary for time-series modeling?
- (3) Which machine learning models are most suitable for short-term bitcoin return prediction, balancing predictive accuracy and model interpretability?

- d. What is the practical value of the proposed forecasting system when applied to real-world trading strategies?

1.3. Research objectives

The primary objective of this study is to develop a multi-source data-driven bitcoin return prediction system on the 4-hour (H4) timeframe by combining statistical analysis, natural language processing, and machine learning techniques. To operationalize this objective, the study pursues the following specific goals. First, to design and implement an automated data collection system from seven heterogeneous sources with a stable ETL architecture. Second, to construct a high-dimensional feature set comprising more than 30 financially motivated and stationary features. Third, to evaluate and select the most suitable machine learning models for financial time-series forecasting. Finally, to assess the practical effectiveness of the proposed system using both statistical metrics (e.g., RMSE, MAE) and finance-specific performance measures (e.g., PnL, Sharpe ratio).

1.4. Scope and limitations

The scope of this study is deliberately defined to focus on bitcoin (BTC), the leading asset in the cryptocurrency market. A 4-hour (H4) timeframe is selected to align with short-term trading objectives. The dataset spans the period from 2018 to 2025 and integrates seven diverse data sources, including OHLCV, market capitalization, network activity, mining metrics, on-chain metrics, news data, and market sentiment indices.

Nevertheless, several limitations should be acknowledged. First, transaction fees and slippage effects associated with large trading volumes are not incorporated into the trading performance evaluation. Second, the modeling framework is restricted to supervised learning methods, and deep learning models are not explored. Finally, the analysis is limited to bitcoin and does not extend to other altcoins.

1.5. Research contributions

This study makes several contributions to the field of financial data science by proposing a complete end-to-end pipeline for cryptocurrency price forecasting, covering all stages from data collection to model evaluation. A key technical contribution lies in the development of a unified framework that integrates five multidimensional feature groups: price action, market structure, on-chain health, sentiment, and inter-market dynamics.

From an empirical perspective, the study demonstrates the superior performance of regularized linear models for short-term (H4) bitcoin return forecasting. From an applied standpoint, the proposed system exhibits strong real-world viability, achieving a substantial positive profit and loss (PnL) of +528% while maintaining effective risk control, with a maximum drawdown of -9%.

2. Theoretical background

2.1. Financial time series and stationarity

Financial time series consist of observations recorded sequentially over time and typically exhibit several distinctive characteristics, including non-stationarity, reflected in time-varying trends and variances; volatility clustering, where periods of high volatility tend to cluster; fat tails, indicating heavier distribution tails than those of a normal distribution; and lagged autocorrelation, where past observations are correlated with current values. In cryptocurrency markets, these properties are even more pronounced due to the absence of traditional stabilizing mechanisms (e.g., central banks), high speculative capital flows, strong reactions to news events, and frequent market regime shifts.

Stationarity is a necessary condition for many statistical and machine learning models. A time series is considered stationary if its mean and variance remain constant over time and its autocovariance depends only on the time lag rather than on the specific time point. To address non-stationarity in price series, this study applies a log return transformation defined as:

$$\log return = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

Log returns remove long-term trends, yield approximately stationary series, exhibit additive properties over time, and tend to follow distributions closer to normality than raw prices, while representing continuous compounded returns with clear economic interpretation.

2.2. Exploratory data analysis

Exploratory data analysis (EDA) is a foundational step in the data analysis pipeline, aiming to understand data structure and distribution, identify patterns, trends, and anomalies, and guide subsequent feature engineering and model selection. In the context of multi-source and multi-frequency cryptocurrency data, EDA plays a particularly important role in distinguishing linear and nonlinear relationships among variables, assessing correlation and multicollinearity, identifying different market regimes (bull, bear, consolidation), and testing the stationarity of time series.

The primary EDA techniques employed in this study include descriptive statistical analysis (mean, median, standard deviation, skewness, kurtosis), visual analysis (time series plots, histograms, boxplots, correlation matrices), hypothesis testing (ADF test for stationarity, Shapiro–Wilk test for normality), and missing data analysis (missingness matrices and pattern analysis).

2.3. Feature engineering

Feature engineering refers to the process of transforming raw data into input variables with clear statistical and financial relevance for machine learning models. In this study, features are constructed across five main groups.

The price action and microstructure group includes both basic and advanced features derived from OHLCV data, such as log return, volume momentum, relative volume (RVOL), Garman–Klass volatility, volatility surprise, signed volatility, close location value (CLV), and wick imbalance. Technical indicators are also incorporated, including Bollinger %B and Bollinger width.

The market structure group focuses on capital flows and macro-level sentiment through features such as stablecoin supply change, stablecoin supply ratio (SSR) as a proxy for buying power, and USDT dominance change as a risk-off indicator.

The on-chain health group leverages blockchain data, including network activity (active address momentum, transaction density), network valuation z-score, mining metrics (hash ribbon, Puell multiple), and on-chain metrics such as the MVRV ratio, exchange netflow, and reserve ratio.

Finally, the sentiment and inter-market group is derived from news and sentiment data, including news sentiment features (impact score, slow decay, shock), the Fear & Greed Index (momentum, divergence relative to MA7), and the rolling 30-day BTC–ETH correlation.

Key feature design principles are strictly followed: all features are required to be stationary, each feature must have clear economic interpretation, shifting techniques are applied to prevent future leakage (rolling statistics are shifted to ensure ex-ante availability), and feature diversity is maintained to cover the five core market dimensions.

2.4. Machine learning for financial time-series forecasting

Machine learning provides powerful tools for modeling complex relationships in time-series data. However, in financial applications, model selection must balance multiple criteria, including predictive accuracy (RMSE, MAE, Spearman IC, directional accuracy), generalization ability, interpretability, and deployability.

The evaluated model families include regularized linear models (ridge, lasso, elastic net), which offer high interpretability, effective overfitting control through regularization, and automatic feature selection in the case of lasso; robust linear models (Huber, SGDRegressor), which are resilient to outliers and scalable to large datasets; support vector machines (LinearSVR and RBF-SVR), which can capture nonlinear patterns but are difficult to interpret; and tree-based ensemble models (decision tree, random forest, gradient boosting), which are effective at modeling nonlinear relationships but are prone to overfitting noise.

Linear models are prioritized in this study for three main reasons: the low signal-to-noise ratio of financial returns makes nonlinear models susceptible to noise overfitting; the engineered features are well-processed (stationary, standardized, and financially meaningful); and regularization effectively controls overfitting while performing implicit feature selection. In addition, high interpretability and fast inference make linear models well suited for real-time trading deployment.

2.5. Model evaluation in a financial context

Unlike standard regression tasks, financial forecasting models must be evaluated using both statistical prediction metrics and trading performance metrics.

Statistical metrics include RMSE (sensitive to outliers), MAE (more robust), Spearman information coefficient (rank correlation between predictions and ground truth), and directional accuracy (the proportion of correctly predicted price directions).

To assess practical value, the study simulates a trading strategy with an initial capital of USD 1 billion and evaluates financial performance using total PnL, win rate, maximum drawdown (MDD), and a Sharpe ratio proxy. The evaluation framework employs a chronological data split (70% training, 15% validation, 15% testing) and a walk-forward simulation, where the test set consists of unseen future data from 2024 to 2025. Performance is benchmarked against a baseline naive momentum strategy.

2.6. Chapter summary

This chapter has presented the research background, research problem, objectives, and theoretical foundations of the study. The key points include the high volatility and 24/7 nature of cryptocurrency markets, the challenge of building a short-term multi-source bitcoin return prediction system, and the theoretical underpinnings related to financial time series, EDA, feature engineering, machine learning, and model evaluation. The subsequent chapters are organized as follows:

- (1) Data collection and preprocessing (Chapter 2)
- (2) Exploratory data analysis and feature engineering (Chapter 3)
- (3) Model development and evaluation (Chapter 4)
- (4) Conclusion (Chapter 5)

CHAPTER II. DATA COLLECTION AND PREPROCESSING

1. Data pipeline

1.1 Overview of the data architecture

The data collection and processing system is designed as a layered ETL (extract–transform–load) architecture consisting of three main layers.

Layer 1: the data collection layer retrieves data from seven different sources at multiple frequencies, specifically 4-hour (H4) intervals for market data and daily (D1) intervals for on-chain data. This layer is implemented with automated updates, intelligent error handling, and resume capability for recovery after interruptions.

Layer 2: the data processing layer integrates the collected data into a master dataset, standardizes timestamps, resolves frequency asynchrony, and performs news sentiment extraction using CryptoBERT. The processed news is then mapped into an actionable timeframe aligned with the trading horizon.

Finally, Layer 3: the feature engineering layer focuses on generating more than 30 features from the seven data groups, ensuring feature stationarity and grounding feature design in quantitative finance principles.

1.2 End-to-end data flow

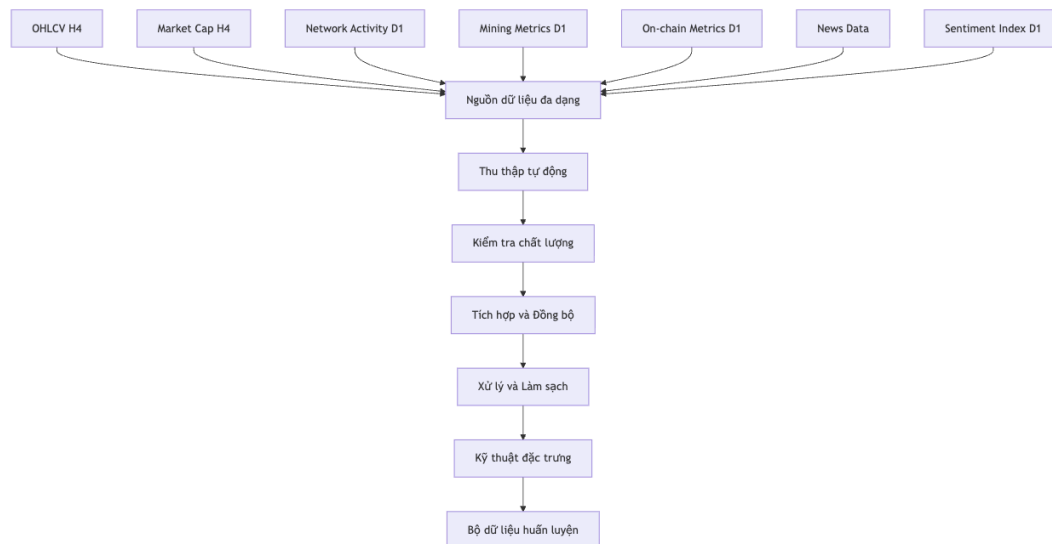


Figure 2.1.1: Overall ETL architecture and data flow diagram

2. Data collection

2.1 Overview of the collection system

The data collection system is implemented under the collectors/ directory, organized into seven independent modules, each responsible for one specific data source. Each

module is designed to be robust and flexible. Key capabilities include restart-from-interruption, where the system detects previously collected data and downloads only missing segments, and intelligent error handling, which automatically retries requests in the presence of network failures or API rate limits. Collected data is stored in a standardized format, using timestamps as the primary key.

2.1.1 Common workflow

All data sources follow a standardized six-step procedure:

- (1) Initialization, including loading configuration and checking existing data
- (2) Determining the time range to be collected
- (3) Fetching data via API calls and downloading in chunks
- (4) Processing data for cleaning, formatting, and merging
- (5) Persisting data with timestamps as the primary key
- (6) Validation to verify data integrity

Notably, the system supports incremental collection via timestamp-based tracking and a merge strategy for duplicate removal. Resume capability ensures the pipeline automatically continues from the last checkpoint after an interruption.

2.1.2 Incremental collection mechanism

Incremental collection is supported through three core mechanisms to ensure continuity and remove duplicates. First, each data source maintains timestamp-based tracking by recording the last successfully collected timestamp. Second, new data is incorporated using a merge strategy, where newly fetched records are merged with historical data and duplicates are removed. Third, resume capability is integrated so the system can automatically continue from the last stopping point in case of failures.

2.1.3 END_TIME configuration and future leakage prevention

The entire collection system enforces a centralized END_TIME configuration to guarantee a consistent observation cutoff and to eliminate the risk of future information leakage during model training and evaluation. This mechanism operates as follows: END_TIME is set globally with a fixed default value; all collected data is automatically truncated to remove observations with timestamps later than END_TIME; incremental updates only fetch data up to END_TIME; and validation checks confirm that no records exceed this cutoff.

This design significantly improves research reproducibility and ensures strict no-leakage conditions.

2.1.4 Overview of data sources

To ensure a multidimensional input dataset, the study integrates seven sources spanning traditional market data, on-chain data, news, and sentiment indicators. Data is collected under different API configurations and frequencies, ranging from 4-hour intervals for trading data to daily or as-published frequencies for on-chain indicators and news.

Table 2.1.1 summarizes the key technical specifications for each source, including collection frequency, number of metrics, API rate limits, chunk size, and the historical time span used in this study.

Source	Frequency	# Metrics	Rate Limit	Chunk Size	Time span
Binance OHLCV	4H	7	1200 req/min	1000 candles	2017-08-17 → 2025-11-14
CoinMarketCap Market Cap	4H	4 coins	~10 req/s	365 days	2013-01-01 → 2025-11-13
CoinMetrics Network Activity	Daily	2 metrics	10 req/6s	365 days	2009-01-03 → 2025-11-14
Blockchain.info Mining	Daily	3 metrics	1 req/s	All-time	2009-01-03 → 2025-11-13
CoinMetrics On-chain Metrics	Daily	8 metrics	10 req/6s	365 days	2009-01-03 → 2025-11-14
Bitcoin Magazine News	As published	N/A	1 req/3s	1 article	2012-02-28 → 2025-11-13
Alternative.me Fear & Greed	Daily	2 fields	N/A	All-time	2018-02-01 → 2025-11-14

Table 2.1.1: Summary of data sources and API configuration

After collection, a total of 58 features are organized by source and data type. Table 2.1.2 provides an overview of feature distribution and clarifies the missing-data strategy applied to each group. These strategies include linear interpolation for price series and forward-fill for lower-frequency metrics.

Source	# Features	Data type	Missing handling	Notes
OHLCV (BTC+ETH)	14	Price, Volume	Linear interpolation	Core market data
Market Cap	4	Market metrics	Forward-fill ($\leq 6h$)	BTC, ETH, USDT, USDC
Network Activity	4	On-chain	Forward-fill	Active addresses, Tx count
Mining	3	Security metrics	Forward-fill	Hash rate, Difficulty, Revenue
On-chain Metrics	8	Valuation	Forward-fill	MVRV, Exchange flows
News Sentiment	23	NLP-derived	Keep NaN	CryptoBERT-based features
Market Sentiment	2	Sentiment index	Forward-fill	Fear & Greed Index
Total	58	Mixed	Strategy-specific	+ 1 timestamp column

Table 2.1.2: Feature categories by source and data type

2.2 Detailed description of data sources

To build a reliable short-term bitcoin (BTC) return prediction system, the study collects and processes data from seven sources to construct a multidimensional feature

set covering market behavior, network activity, sentiment, and macro-level information.

2.2.1 Price data (OHLCV)

OHLCV data (open, high, low, close, volume) for the BTC/USDT and ETH/USDT trading pairs is retrieved from the Binance API at a 4-hour (H4) frequency, covering market history since 2017.

The collection procedure is designed to preserve time-series continuity and prevent future leakage. Data is downloaded in chunks of 1000 candles, carefully merged, and truncated using the predefined end timestamp. Small data gaps are filled using linear interpolation to maintain continuity, while missing volume values are handled via forward-fill. The system also includes automatic interruption detection and resume-from-checkpoint functionality, improving ETL stability.

2.2.2 Market capitalization data

Market capitalization data is collected from the CoinMarketCap API, focusing on major assets and widely used stablecoins such as BTC, ETH, USDT, and USDC. Due to API constraints, data is retrieved in 365-day chunks. Assets are merged using an outer join to retain information despite asynchronous update schedules. Time misalignment is normalized through forward-fill with a maximum limit of 6 hours, preserving consistency without distorting long-term macro trends.

2.2.3 Network activity data

Core on-chain indicators, including active addresses and transaction count for both Bitcoin and Ethereum, are retrieved from the CoinMetrics Community API at a daily frequency. To ensure stability and policy compliance, the pipeline strictly enforces the 6-second rate limit and includes retry logic to handle connection errors, ensuring reliable inputs for network-usage features.

2.2.4 Mining and security data

Mining and security indicators such as hash rate, mining difficulty, and miner revenue are collected from the Blockchain.info API. These metrics reflect network security and production costs in the Bitcoin ecosystem. The dataset is transformed from long format to wide format using pivot operations to optimize multivariate time-series analysis. Missing values are handled via forward-fill, and outliers are checked and adjusted using statistical thresholds to improve data quality.

2.2.5 Advanced on-chain metrics

Advanced on-chain indicators (e.g., MVRV ratio, exchange flows, and circulating supply metrics) are collected from the CoinMetrics API in 30-day chunks. These metrics provide deeper signals regarding long-term positioning, whale behavior, and relative valuation. Consistent with other sources, fields are mapped into standardized naming conventions and merged with historical data to ensure continuity.

2.2.6 Bitcoin Magazine news data

Textual news data is collected from the official Bitcoin Magazine sitemap, including raw HTML content and metadata (URL, publication date), spanning 2012 to 2025. A robust crawling pipeline with state tracking and resume capability is used to handle large-scale data collection, resulting in 13,388 articles over nearly 14 years. The annual distribution is summarized in Table 2.1.3.

Year	# articles	Notes
2012	79	Early stage
2013	233	
2014	604	
2015	702	Rapid growth
2016	503	
2017	519	Bull run
2018	802	Peak activity
2019	493	
2020	165	Sharp decline
2021	1,197	Bull run, strong growth
2022	1,191	Sustained high volume
2023	659	
2024	778	
2025	5,463	Sharp increase (up to 11/2025)
Total	13,388	

Table 2.2.1: Total number of articles collected from Bitcoin Magazine by year

Using multiple CSS selectors with priority ordering improves extraction robustness even when the website’s HTML structure changes over time.

2.2.7 Market sentiment index data

To represent short-term investor sentiment, the Fear & Greed Index is collected daily from Alternative.me. The collector is designed with fallback mechanisms and adaptability to potential API structure changes, ensuring continuity for downstream sentiment feature construction.

2.3 News processing and sentiment analysis

2.3.1 Article content extraction and normalization

News processing begins with extracting and cleaning textual content from raw HTML. The system parses HTML to retrieve key metadata such as title and author. A strict cleaning step removes irrelevant technical elements (script, style, iframe) and

advertisement blocks. Main content is extracted using intelligent paragraph detection and filtered using a minimum length threshold (15 characters) to improve quality and reduce noise. This pipeline is designed to remain stable across layout changes over the long study period (2012–2025).

2.3.2 Domain-specific sentiment modeling with CryptoBERT

To quantify market sentiment, the domain-specific language model CryptoBERT is applied. Since news articles often exceed the model’s input length limit, an effective token-based chunking strategy is used. Sentiment analysis is performed at two levels: head-focused sentiment, which combines the headline with the beginning of the article (~1000 characters) to approximate a reader’s first impression, and global sentiment, which analyzes the full article to capture the overall narrative.

In addition, to detect extreme events, the pipeline extracts top-k extreme sentiment signals. This step produces a set of 11 multidimensional sentiment features, including mean bullish and bearish probabilities, maximum values, and a sentiment net score, providing richer signals for the prediction model.

2.3.3 Mapping news to an actionable timeframe

To integrate sentiment-analyzed news with the H4 market time series, a strict as-of mapping mechanism is implemented to eliminate future leakage. Under this principle, an article is assigned only to the candle where it is actually available for trading decisions. Specifically, news published within the interval $(T-\Delta, T]$ is mapped to the opening time of the next candle T . This ensures that the model only uses information that would have been available at prediction time. The design is general and can be extended to other timeframes.

2.4 Master data integration

The integration stage merges heterogeneous data sources into a unified master dataset, synchronized along the time axis and prepared for subsequent analysis and modeling.

First, all sources are loaded and timestamps are standardized to ensure consistency in timezone and temporal resolution. News data, after extraction and sentiment processing, is aggregated over time and incorporated into the merging workflow.

The merge is performed using an outer join on the timestamp field to preserve as much information as possible, even under differences in update frequency or coverage periods. After merging, the system reindexes the timeline by creating a complete H4 time grid across the full research horizon, ensuring continuity and enabling consistent model training.

Source	# records	Size	Time span
OHLCV (BTC)	18,050	1.65 MB	2017-08-17 → 2025-11-14
OHLCV (ETH)	18,050	1.55 MB	2017-08-17 → 2025-11-14
Market Cap	17,129	1.35 MB	2013-01-01 → 2025-11-13
Network Activity (BTC)	6,160	257 KB	2009-01-03 → 2025-11-14
Network Activity (ETH)	3,761	162 KB	2015-07-30 → 2025-11-14
Mining Metrics	3,077	228 KB	2009-01-03 → 2025-11-13
On-chain Metrics	6,160	868 KB	2009-01-03 → 2025-11-14
News	13,388	1.62 MB	2012-02-28 → 2025-11-13
Sentiment Index	2,840	103 KB	2018-02-01 → 2025-11-14
Master Dataset	36,955	8.46 MB	2009-01-03 → 2025-11-14

Table 2.2.2: Overview of collected datasets

The final master dataset contains 60 feature columns, distributed as follows. Bitcoin-related features contribute 18 columns, including 7 OHLCV fields (open, high, low, close, volume, quote volume, trades), 1 market capitalization field, 2 network activity fields (active addresses, transaction count), 3 mining indicators (hash rate, difficulty, miner revenue), and 8 on-chain metrics (total supply, MVRV ratio, exchange flows, exchange supply). Ethereum-related features contribute 10 columns, including 7 OHLCV fields, 1 market capitalization field, and 2 network activity fields. Stablecoin variables are represented by 2 columns (USDT and USDC market cap). News sentiment contributes the largest share with 23 columns, including news volume indicators, combined sentiment statistics, bull/bear structure features, CryptoBERT

probability outputs, and a conflict detection signal. Finally, the market sentiment index (Fear & Greed Index) is integrated as well.

Overall, the dataset contains 36,955 samples (approximately 16.9 years of continuous H4 intervals) with a CSV size of 8.46 MB. It should be noted that missingness varies across feature groups, particularly for news sentiment features, since news does not occur in every H4 window.

2.5 Challenges and solutions

During the design and operation of the large-scale multi-source data pipeline (over 13 years of history and high-frequency collection), several critical technical challenges were identified and addressed. The implemented solutions aim to ensure data integrity, processing efficiency, and pipeline resilience.

2.5.1 API rate limiting

A major challenge in collecting price and on-chain data is API rate limiting imposed by providers such as CoinMetrics and Binance. Exceeding these limits typically results in HTTP 429 errors and may lead to temporary access bans. To mitigate this, a multi-layer traffic control strategy is implemented.

First, a chunking strategy is applied, where historical data is split into smaller blocks (e.g., 365 days per chunk) to reduce the risk of large-scale interruptions. Second, source-specific delays are enforced (e.g., 0.1s for Binance, 2s for CoinMarketCap, and 6s for CoinMetrics) to comply strictly with access policies. Third, an exponential backoff mechanism is used when HTTP 429 errors are detected, allowing the pipeline to pause, gradually increase wait times, and retry requests, significantly improving end-to-end robustness.

2.5.2 Missing data and data gaps

Multi-source and multi-frequency datasets are inherently exposed to missingness due to API downtime, schema changes, special network events (e.g., hard forks), or natural differences in update cadence. The system therefore applies flexible imputation strategies. Most gaps are handled using forward-fill (LOCF), which carries forward the most recent observed value. For OHLCV price data, linear interpolation is preferred for short gaps to preserve continuity. During multi-source integration, outer joins are used to retain all available information. Large gaps that cannot be logically interpolated are kept as null values and handled later during feature engineering.

2.5.3 Future Leakage Prevention

Future leakage is a critical risk in financial time-series forecasting, often arising when data is updated after decision time or when timestamps are misaligned across sources. The primary safeguard is a strict `END_TIME` cutoff that removes any observations beyond the defined horizon.

In particular, an as-of news mapping rule is enforced: news published in $(T-\Delta, T]$ is assigned to the candle opening at time T , ensuring the model only uses information available at prediction time. In addition, timestamp validity checks and no-lookahead aggregation rules are applied so that every engineered feature is computed exclusively from data available up to time T .

2.5.4 HTML Layout Changes (2012-2025)

For long-horizon news data (2012–2025), frequent changes in HTML layout and CSS selectors create a significant extraction challenge. The pipeline addresses this by using multiple selectors with priority ordering to adapt across website versions. It also applies recursive parsing with robust paragraph detection rather than relying on specific HTML tags. Only paragraphs above a minimum length threshold are retained to reduce noise. If extraction fails, the system returns empty content via graceful degradation rather than raising errors, preserving pipeline stability.

2.5.5 Incremental collection and resume capability

Collecting long historical data is time-consuming and prone to interruptions caused by network issues or system failures. To improve efficiency, a checkpoint mechanism records progress after each completed chunk. The system supports timestamp-based resume by reading the last stored timestamp and fetching only new data thereafter. This is combined with a merge strategy that removes duplicates and ensures idempotent operations, allowing scripts to be safely re-run without generating redundant records.

3. Data cleaning

3.1 Overview of the data cleaning process

The data cleaning pipeline is designed to ensure that the input dataset satisfies key requirements for machine learning modeling, including temporal continuity, logical consistency across variables, and uniform usability when integrating heterogeneous data sources. Given the nature of time-series data in crypto markets (price–volume, on-chain indicators, sentiment indices, and news data), the cleaning pipeline focuses on addressing four core issues:

- (1) Missing data due to API limitations or source misalignment
- (2) Extreme values that may adversely affect model training
- (3) Logical constraint checks for financial variables (especially OHLC)
- (4) Standardizing handling strategies based on update frequency (H4 vs D1) and the economic nature of each feature

The process begins with a comprehensive assessment of missingness in the master dataset. Visualization via a missingness matrix indicates that missing data are not random; instead, they exhibit clear “availability milestones” by source group: OHLCV

data start from 2017-08-17; the sentiment index starts from 2018-02-01; USDC market capitalization appears only from 2018-10-09 because the asset did not exist prior to that date; meanwhile, news data are available much earlier (2012-02-28) but remain highly sparse, reflecting the inherently discontinuous nature of news at the hourly level. Correctly identifying these missingness patterns is essential for selecting appropriate handling strategies and avoiding the introduction of spurious signals in financial data.

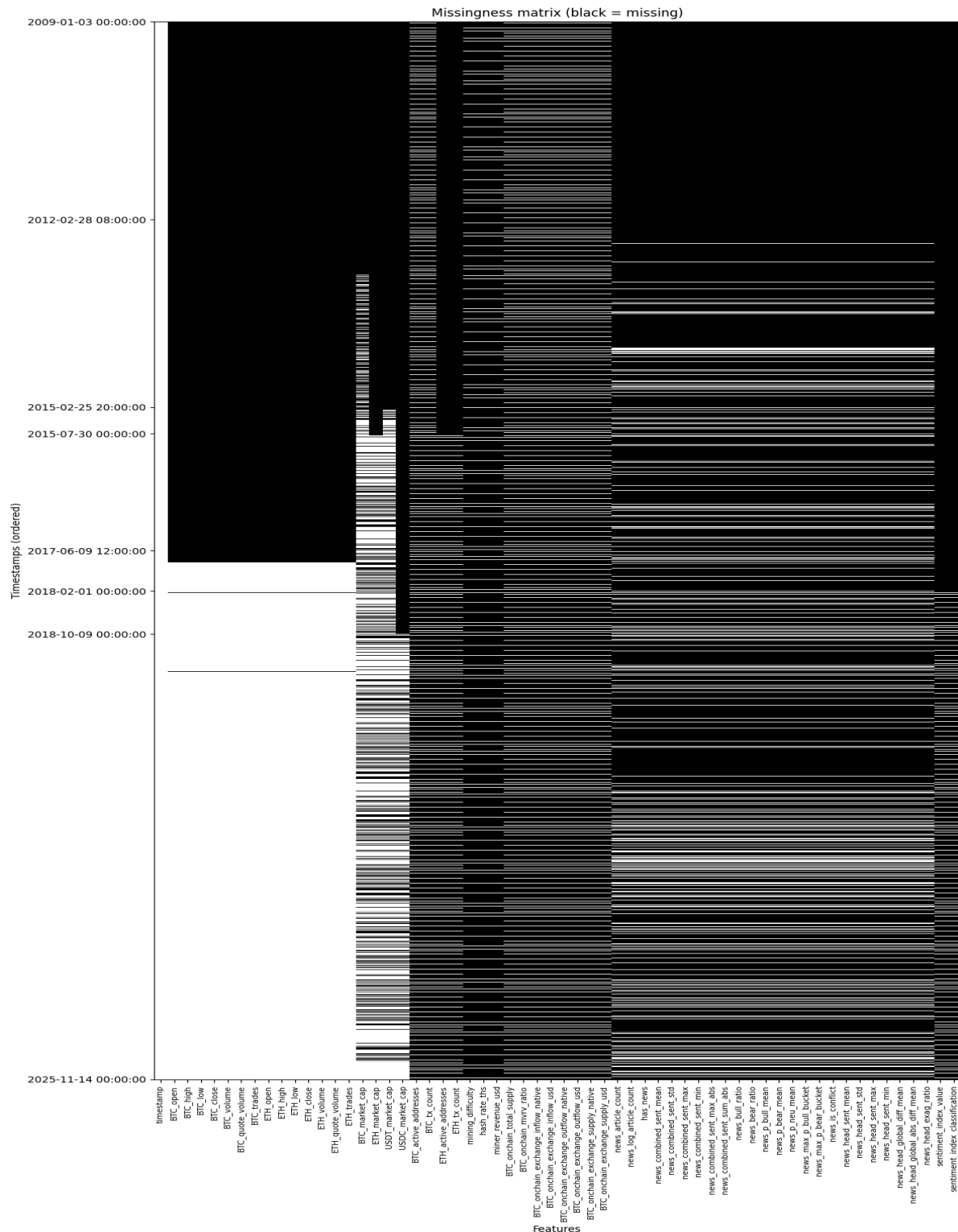


Figure 2.3.1. Missingness matrix of the master dataset

3.2 Determining a valid start date

Because the dataset is aggregated from multiple sources with different availability dates, selecting the start date is not only a technical decision (reducing missingness) but also an economic and statistical one (avoiding inappropriate interpolation that distorts market behavior). In this study, the guiding principle is that the dataset start date should be determined by feature groups that should not be interpolated; otherwise, the system would be forced to impute values for variables where imputation is economically unjustified.

3.2.1 Rationale for selecting the start date

In practice, bitcoin prices exhibit well-known financial time-series properties such as random-walk-like behavior, volatility clustering, and fat tails. As a result, interpolating price data—especially over long gaps or across periods where data are missing because they did not exist—can introduce artificial smoothness and distort the return distribution. Therefore, the start date should not be based on sources that are “available earlier but sparse” (e.g., news); instead, it must be anchored to the data group that serves as the foundation for feature engineering, namely OHLCV.

Based on this rationale, the dataset start date cannot precede the earliest OHLCV availability. Choosing an earlier start would effectively require “fabricating price data” through imputation, which violates the principle of preserving true market structure.

3.2.2 Xác định mốc thời gian tối ưu cho tập master

Although OHLCV is foundational, the master dataset requires a start date that ensures adequate coverage across the most critical feature groups. Among the data sources, the sentiment index is decisive because missingness prior to 2018-02-01 is not primarily due to collection errors, but rather due to provider limitations and data accessibility constraints. Moreover, since sentiment is a behavioral market variable, interpolation is likely to generate artificial sentiment signals. For these reasons, 2018-02-01 is selected as the final start date for the cleaned dataset.

In contrast, missingness in USDC market capitalization before 2018-10-09 reflects the fact that USDC had not been launched. In this case, assigning a value of 0 prior to launch is economically consistent (market cap = 0 when the asset does not exist) and does not introduce spurious signals.

3.3 Selecting the optimal start date for the master dataset

After finalizing the start date and the structure of the master dataset, variables are categorized by update frequency and economic nature to select appropriate imputation strategies. This approach ensures that “filling” missing values remains consistent with how the data are generated in real-world settings.

3.3.1 H4 data group: OHLCV and market capitalization

The H4 group includes BTC/ETH OHLCV data and market capitalization data for BTC, ETH, USDT, and USDC. For OHLCV, missingness mainly arises from two causes:

- (1) The master timeline starts earlier than the OHLCV availability
- (2) Short disruptions from the API

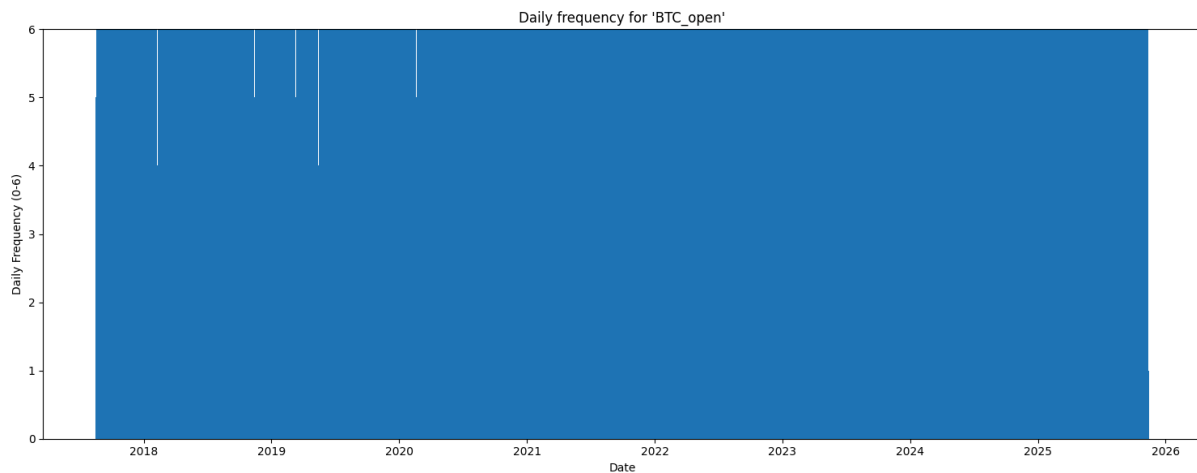


Figure 2.3.2. Daily data availability frequency for the OHLCV group

For OHLCV features collected via the Binance API, missingness tends to be short and intermittent. Therefore, linear interpolation is applied to price and volume variables to preserve temporal continuity without distorting trends over very short intervals.

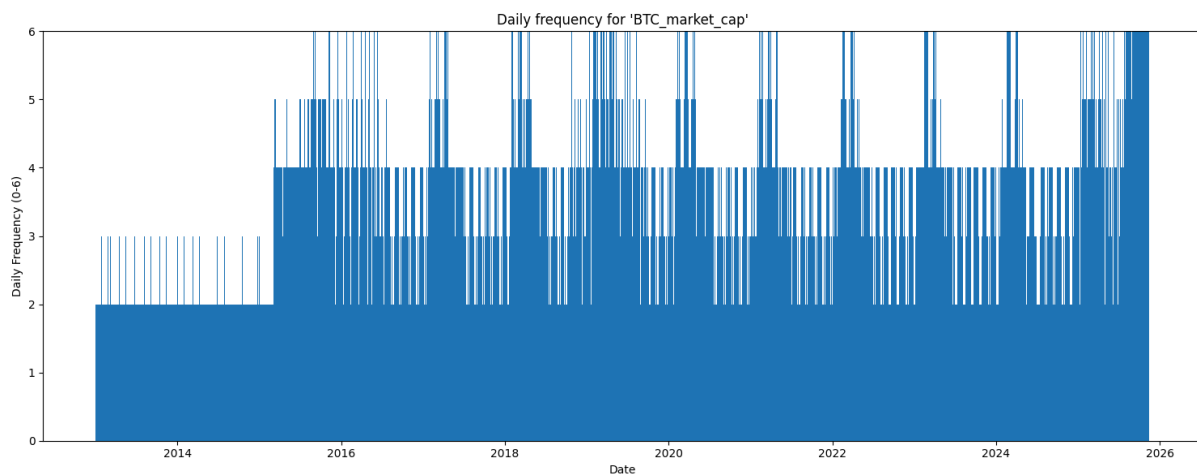


Figure 2.3.3. Daily data availability frequency for BTC market capitalization

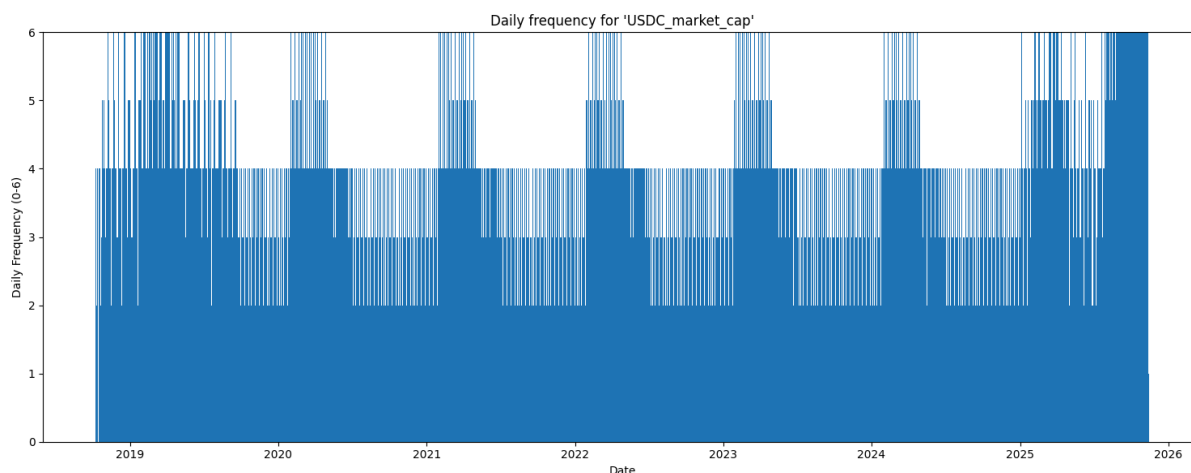


Figure 2.3.4. Daily data availability frequency for USDC market capitalization

For market capitalization, missingness is typically higher and uneven across assets due to differences in historical coverage and update mechanisms. Since missingness is often prolonged and reflects asynchronous updates, linear interpolation is no longer appropriate. Instead, the study applies forward-fill (LOCF) to carry the most recent value forward until a new observation becomes available, while periods “prior to existence” are handled with a value of 0 (notably for USDC).

3.3.2 News data (news features)

The news group includes features capturing news intensity, sentiment distribution, and embedding-based representations (CryptoBERT) on the H4 timeframe. Unlike market data, missingness in news is not necessarily an “error”; in many cases, it simply reflects that no article was published during that 4-hour window. Therefore, missing values after aggregation are interpreted as an absence of information rather than technical missingness.

Accordingly, default values are assigned for windows with no news: count-based variables are set to 0, boolean variables are set to False, and an additional flag variable `has_news` is introduced to distinguish between two important cases: (i) no news, and (ii) news present but neutral or low impact. This prevents the model from conflating “no observed news” with “news impact equals zero.”

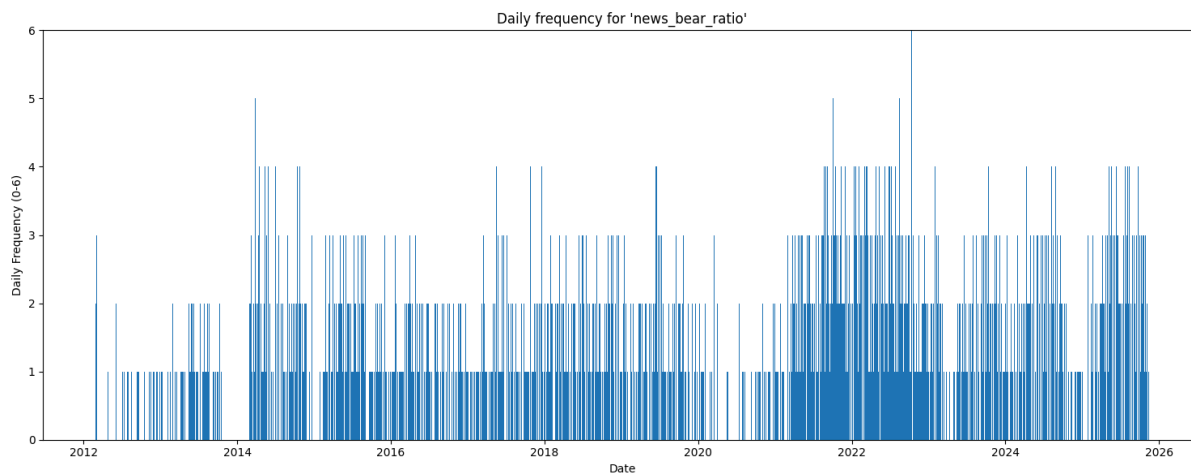


Figure 2.3.5. Daily news occurrence frequency (number of H4 windows with news)

3.3.3 D1 data group (network, mining, and on-chain)

The D1 group includes daily network activity indicators, mining metrics, on-chain metrics, and selected sentiment indicators. When mapped to the H4 timeframe, null values in D1-derived columns are a natural consequence of daily data availability (and mining metrics may update every 2–4 days). In this setting, forward-fill (LOCF) is a standard and appropriate choice: the most recent value is carried across subsequent H4 windows until the next daily observation becomes available. This approach avoids creating artificial information while still providing complete features for modeling.

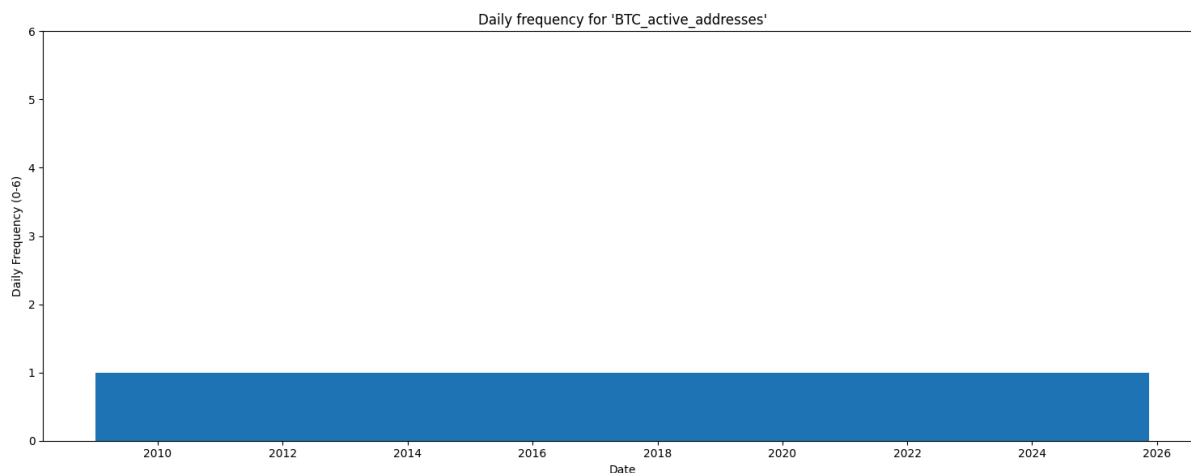


Figure 2.3.6. Daily update frequency of the D1 group (excluding mining)

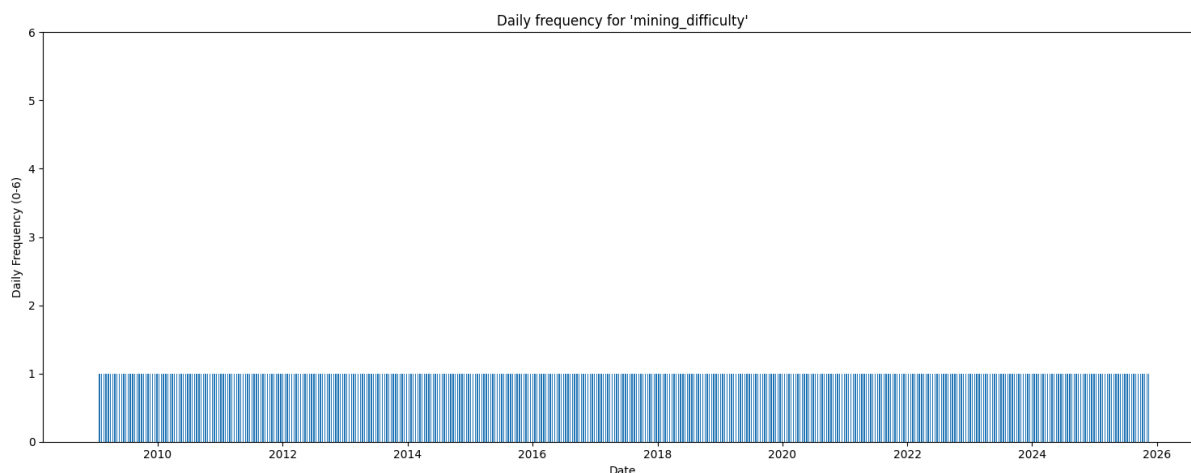


Figure 2.3.7. Daily update frequency of the mining group

3.4 Data quality checks after cleaning

After applying group-specific handling strategies, the dataset is validated through a set of quality checks to ensure reliability prior to modeling.

3.4.1 Missingness after processing and start-date filtering

The cleaned dataset is filtered to start from 2018-02-01 (the first date of sentiment index availability) to ensure that all features are present throughout the analysis period. Results show that the H4 series becomes continuous, and the remaining missingness is negligible ($<0.01\%$), primarily occurring in edge cases.

3.4.2 Negative value checks

Negative value checks are conducted to ensure there are no values that violate economic logic after cleaning. The analysis shows that negative values appear only in the sentiment feature group derived from news data, including aggregated statistics such as the minimum, mean, and maximum sentiment scores, as well as variables measuring the difference between headline-level sentiment and overall sentiment.

These negative values are economically valid: sentiment scores can be negative under bearish news and positive under bullish news. For difference-based variables, negative values are expected when headline sentiment is lower than overall sentiment within the same time window. Therefore, negative values in these features reflect the true nature of the data rather than collection or processing errors.

In contrast, strictly non-negative quantitative variables such as price, trading volume, and market capitalization contain no negative values, indicating that the economic logic constraints have been preserved throughout the cleaning process. Overall, all negative values in the dataset are valid and economically interpretable, and no additional adjustments are required prior to modeling.

3.4.3 Infinite value checks

No infinite values ($\pm\infty$) are detected in the dataset. This ensures numerical stability during optimization and subsequent transformations.

3.4.4 OHLC logical consistency checks

OHLC constraints are validated to ensure consistency: low does not exceed open/close/high, and high is greater than or equal to all other price levels. No violations are found for either BTC or ETH, indicating good price data quality after cleaning.

3.4.5 Outlier analysis and handling decisions

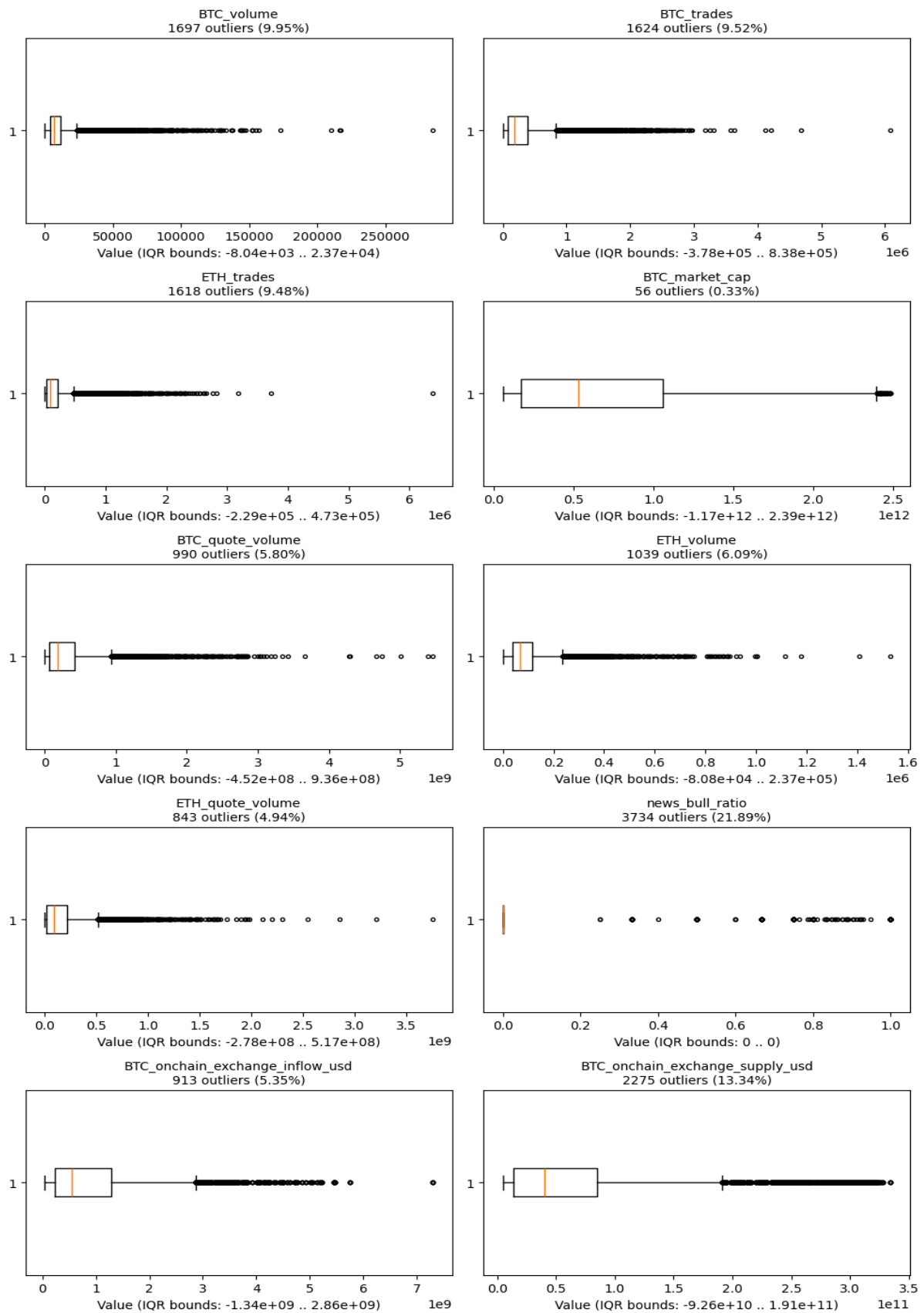


Figure 2.3.8. Boxplots of variables with the highest outlier rates

Outliers are detected using the IQR method with a $1.5 \times \text{IQR}$ threshold. Variables with large numbers of outliers are concentrated in volume, transaction count, market capitalization, and selected news-related features—an expected phenomenon in crypto markets due to high volatility and abrupt events. Outliers constitute a small fraction of total data points, and many extreme observations may represent genuine market events (e.g., pump/dump episodes, major news, regulatory shocks). Therefore, outliers are retained to preserve information and reflect the heavy-tailed nature of crypto data, rather than “smoothing” the dataset in a way that could remove meaningful signals.

3.5 Results after data cleaning

After completing the cleaning process and quality checks, the dataset meets key modeling requirements: a continuous H4 time series, preserved OHLC logical constraints, no infinite values, economically valid negative values confined to sentiment features, and outliers retained to reflect the true distributional characteristics of crypto markets.

Indicator	Value
Number of records	17,059
Number of features	60 (including timestamp)
Time span	2018-02-01 to 2025-11-14

Table 2.3.1. Summary statistics of the cleaned dataset

Statistically, the final dataset contains over 17,000 observations, 60 features (including timestamp), and covers 2018-02-01 to 2025-11-14, with 0% missingness and no OHLC violations. The dataset is stored in two formats: Pickle to optimize processing speed and preserve Python data types, and CSV for inspection, sharing, and compatibility with other analysis tools.

4. Exploratory data analysis and feature engineering

4.1 Overview of the cleaned dataset

After completing the data cleaning process in Chapter 3, the master dataset contains approximately 17,000 observations at a 4-hour (H4) resolution, covering the period from 2018-02-01 to 2025-11-14. The dataset includes 60 raw variables (including the timestamp) and contains no missing values. Variables are collected from seven main

data groups, providing comprehensive coverage of key dimensions of the cryptocurrency market:

- (1) Market data (BTC/ETH OHLCV)
- (2) Market capitalization (BTC, ETH, and stablecoins)
- (3) Network activity
- (4) Mining indicators
- (5) On-chain metrics for valuation and exchange inflow/outflow dynamics
- (6) News data
- (7) Market sentiment index (Fear & Greed)

Based on the cleaned dataset, this chapter focuses on two core tasks:

- (1) Conducting exploratory data analysis (EDA) to understand the distributional properties and dynamics of each variable group
- (2) Performing feature engineering to transform raw variables into stationary, economically meaningful quantities suitable for machine learning models that forecast short-term bitcoin returns

A detailed list of raw variables and their descriptions is provided in Appendix A.

4.2 OHLCV analysis and feature engineering

4.2.1 Target construction and stationarity testing

In financial markets, price series are typically non-stationary due to long-term trends and time-varying variance. Using raw prices directly in machine learning models can lead to learning spurious trends or being dominated by scale changes. Therefore, the first step in feature engineering is to transform price data into stationary variables with clear economic meaning, namely log returns.

The target variable in this study is the future BTC return of the next 4-hour candle, defined by shifting the log return one step forward:

$$BTC \log return_t = \ln\left(\frac{BTC \text{ close}_t}{BTC \text{ close}_{t-1}}\right)$$
$$target = BTC \log return_{t+1}$$

This definition ensures that at time t , the model uses information available up to t to predict the return of the next period ($t+1$), which aligns with a short-term trading setting.

Augmented Dickey–Fuller (ADF) test results show that the `BTC_close` series fails to reject the unit root hypothesis ($p\text{-value} > 0.05$), while `BTC_log_return` rejects it at the 5% significance level ($p\text{-value} < 0.05$). This confirms that log-differencing transforms the price series into a stationary form, which is important for modeling and for reducing the risk of spurious correlation.

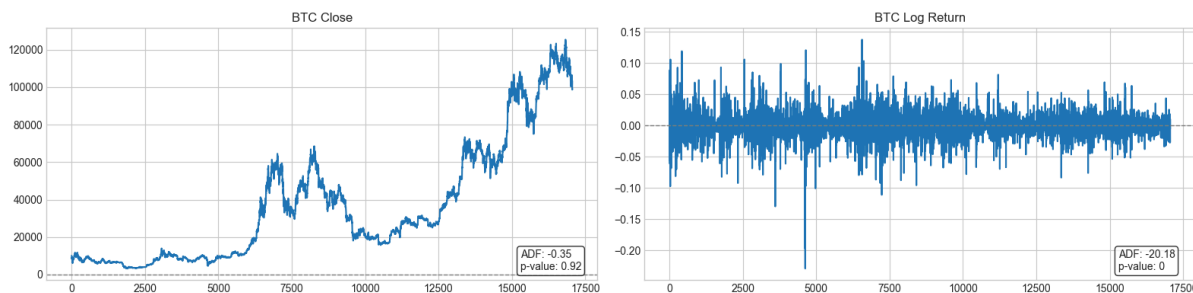


Figure 2.4.1. Comparison of BTC close price and BTC log return

4.2.2. Return distribution analysis

Beyond stationarity, the return distribution is a critical financial characteristic because it determines the severity of shocks and the likelihood of extreme events. Comparing the distributions of close prices and log returns shows that close prices are right-skewed, exhibit multiple modes across market regimes, and undergo strong scale shifts over time. In contrast, log returns are concentrated around zero and are more symmetric, but still exhibit heavy tails.

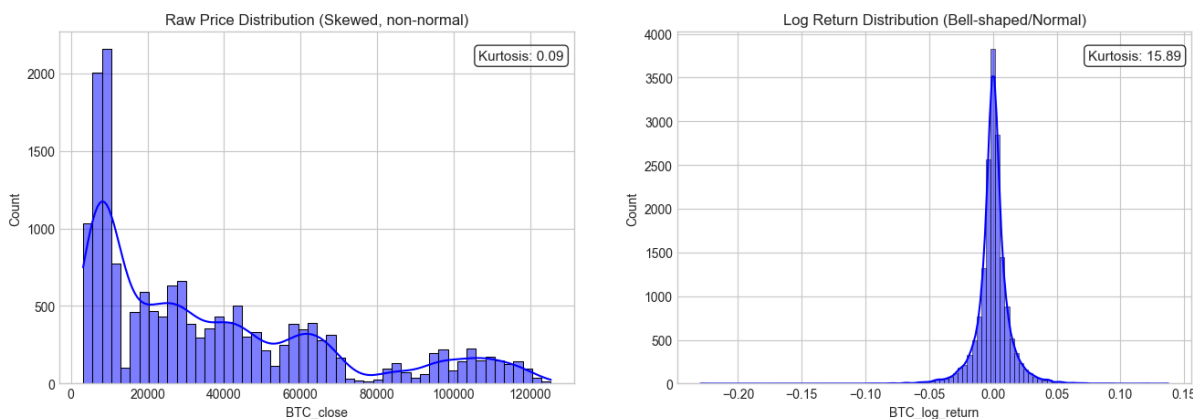


Figure 2.4.2. Distributions of BTC close price and BTC log return

The sample kurtosis of log returns is approximately 15.89, indicating a substantially higher probability of extreme price movements compared to a normal distribution. This is a well-known property of financial time series and is particularly pronounced in crypto markets, where crashes and rallies can occur abruptly. Accordingly, the modeling stage should prioritize models and regularization strategies that are robust to noise and do not rely on idealized normality assumptions.

4.2.3 Retained features

Three variables are retained from the OHLCV group. First, the current bitcoin log return is kept as a stationary representation of recent price movement and serves as the core state variable for short-horizon prediction. Second, the ethereum log return is retained to capture inter-market co-movement and cross-asset spillovers that are common in crypto markets. Third, the next-period bitcoin log return is used as the target variable, ensuring an ex-ante forecasting setup where only information available up to time t is used to predict $t+1$.

4.3 Volume analysis and feature engineering

Unlike price, trading volume reflects market participation and the intensity of capital flows. However, raw volume is often non-stationary, contains many outliers, and is strongly affected by market scale over time, making absolute levels noisy. Therefore, volume features are designed to capture momentum and abnormality relative to a baseline.

4.3.1 Volume momentum (volume log return)

Trading volume captures participation intensity, but raw volume is typically non-stationary due to market growth and is prone to outliers. In addition, absolute volume depends on price levels and market microstructure across periods, making direct comparisons across cycles less meaningful.

To address this, the study computes volume momentum by applying a log transform (to stabilize the distribution and handle zeros) and taking differences between consecutive periods. Empirically, the transformed series oscillates around zero and is substantially more stable than raw volume.

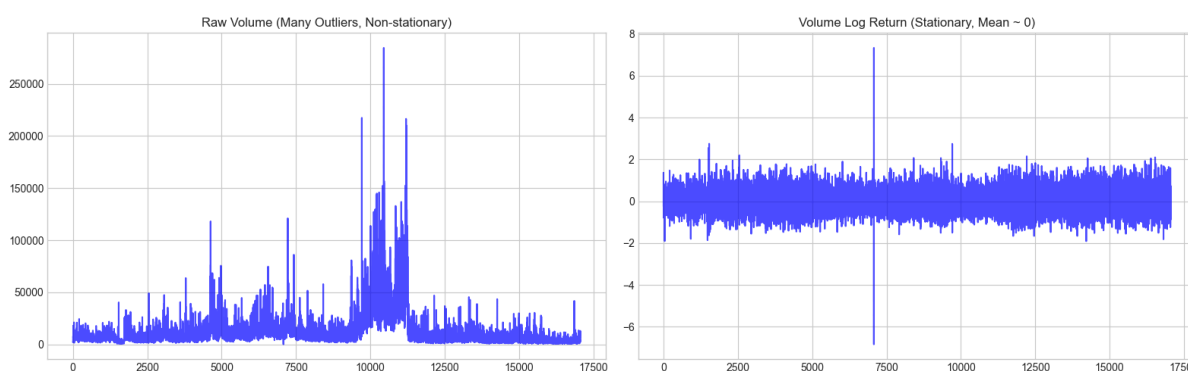


Figure 2.4.3. Comparison of BTC raw volume and transformed volume momentum

4.3.2 Relative Volume (RVOL)

Volume momentum captures short-term changes but does not indicate whether current volume is high or low relative to its baseline. Therefore, relative volume (RVOL) is constructed by normalizing current volume by the 20-period moving average. To prevent future leakage, this moving average is shifted one step backward (ex-ante). RVOL is then log-transformed to reduce right-skewness and make the distribution

more symmetric, while still preserving heavy-tailed characteristics consistent with financial data.

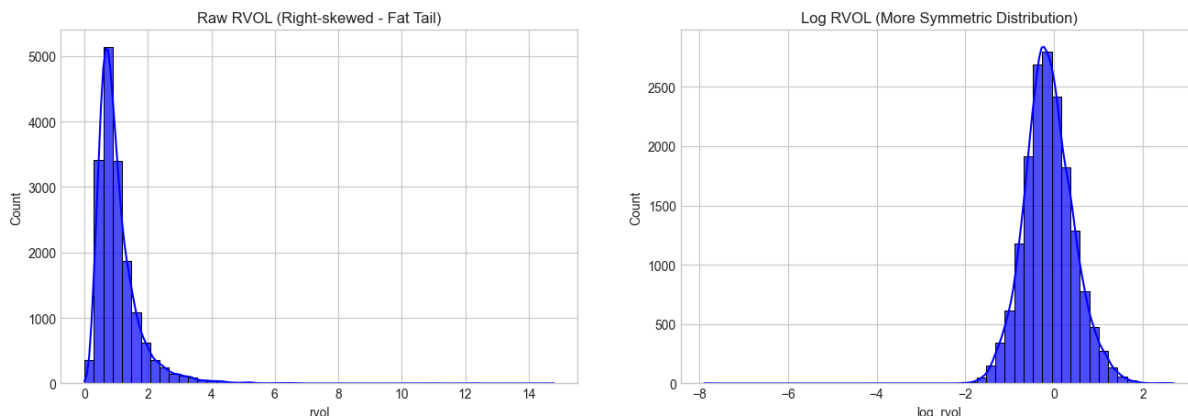


Figure 2.4.4. Distributions of RVOL and $\log(\text{RVOL})$

4.3.3. Retained features

Two volume-derived features are retained. The volume momentum (log-volume change) is kept because it captures abrupt changes in trading activity in a stationary form, which often accompanies short-term regime shifts. In addition, relative volume (log-normalized RVOL) is retained because it measures whether current participation is unusually high or low relative to a recent baseline, providing complementary information beyond instantaneous volume changes. Both transformations reduce scale effects and mitigate the instability of raw volume.

4.4 Phân tích và kỹ thuật đặc trưng Volatility

The volume group retains two variables: volume momentum and $\log(\text{RVOL})$, representing flow momentum and flow abnormality relative to baseline, respectively.

4.4.1 Garman–Klass volatility: Realized volatility from OHLC

Instead of using close-to-close volatility, the study estimates realized volatility from OHLC using the Garman–Klass method, which leverages intrabar high/low information to better reflect within-candle variation. Empirical observations show clear “expansion” phases during turbulent markets and “contraction” phases during range-bound periods.

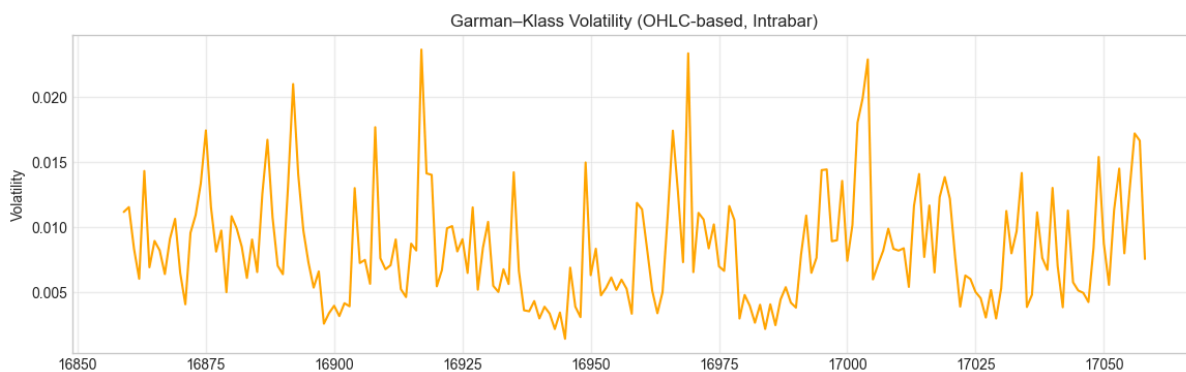


Figure 2.4.5. Garman–Klass volatility in a zoomed window, highlighting expansion and contraction phases

4.4.2 Volatility surprise (regime-normalized)

Because volatility clusters, absolute volatility levels are insufficient for identifying abnormal regimes. The study normalizes volatility using a z-score relative to the rolling mean and standard deviation over 50 periods. This enables the model to detect volatility shocks via thresholds (e.g., beyond ± 3 standard deviations) and to distinguish regime-normal volatility from abnormal volatility.

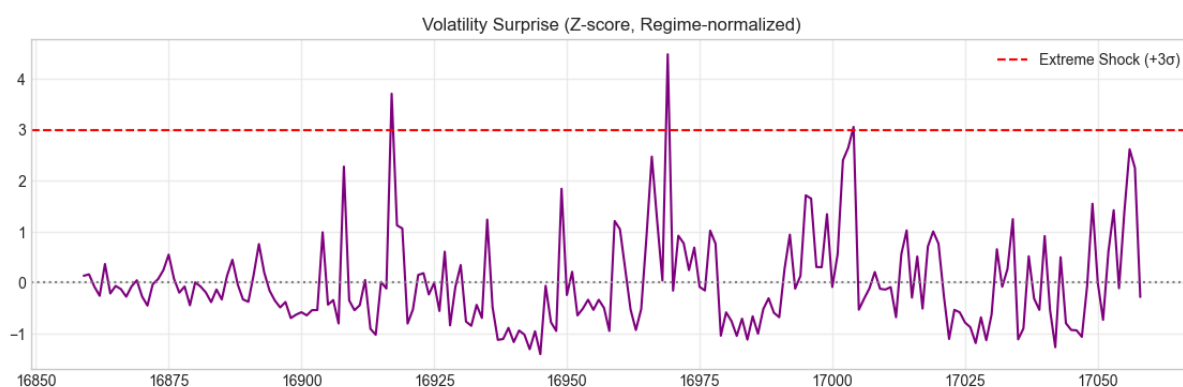


Figure 2.4.6. Volatility surprise with $\pm 3\sigma$ thresholds for detecting regime shocks and squeezes

4.4.3 Signed volatility (directional volatility)

Standard volatility is non-negative and cannot distinguish volatility associated with buying pressure versus selling pressure. To address this, volatility is signed using the direction of the same-period log return, producing signed volatility. This feature helps differentiate “volatility with upward moves” from “volatility with downward moves,” which is meaningful in highly volatile crypto markets.

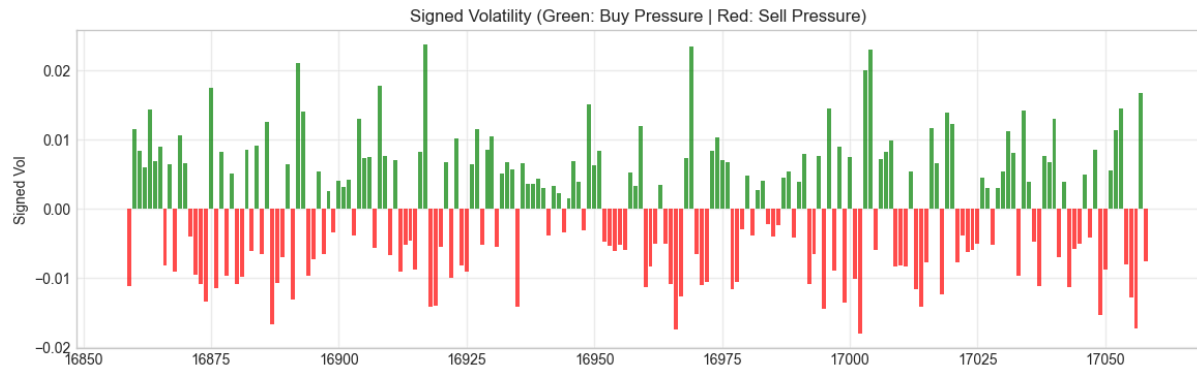


Figure 2.4.7. Signed volatility over time, showing differences between volatility during up moves and down moves

4.4.4 Retained features

The volatility group retains three features that represent risk conditions at different levels. Realized intrabar volatility (Garman–Klass) is kept because it extracts more information from OHLC data than close-to-close measures and better reflects within-candle uncertainty. Regime-normalized volatility (volatility surprise) is retained to highlight volatility shocks relative to recent conditions rather than absolute levels, which is essential under volatility clustering. Finally, directional (signed) volatility is included to distinguish volatility associated with upward versus downward moves, which can be informative in a market where sell-offs and rallies are asymmetric.

4.5 Microstructure analysis and feature engineering

Microstructure analysis focuses on how price moves within a candle—where it closes relative to the range and which side the wick extends—providing signals about market participant behavior.

4.5.1 Close location value (CLV): close position within the candle

CLV measures the relative position of the close price within the high–low range, reflecting whether buyers or sellers dominate into the end of the period. A close near the high is typically interpreted as sustained buying strength, whereas a close near the low suggests seller dominance.

To evaluate usefulness, CLV is bucketed into five levels (from strongly bearish to strongly bullish), and next-period returns are compared across buckets. The results show structured differences in next-candle returns, suggesting that CLV captures behavioral signal rather than pure noise.

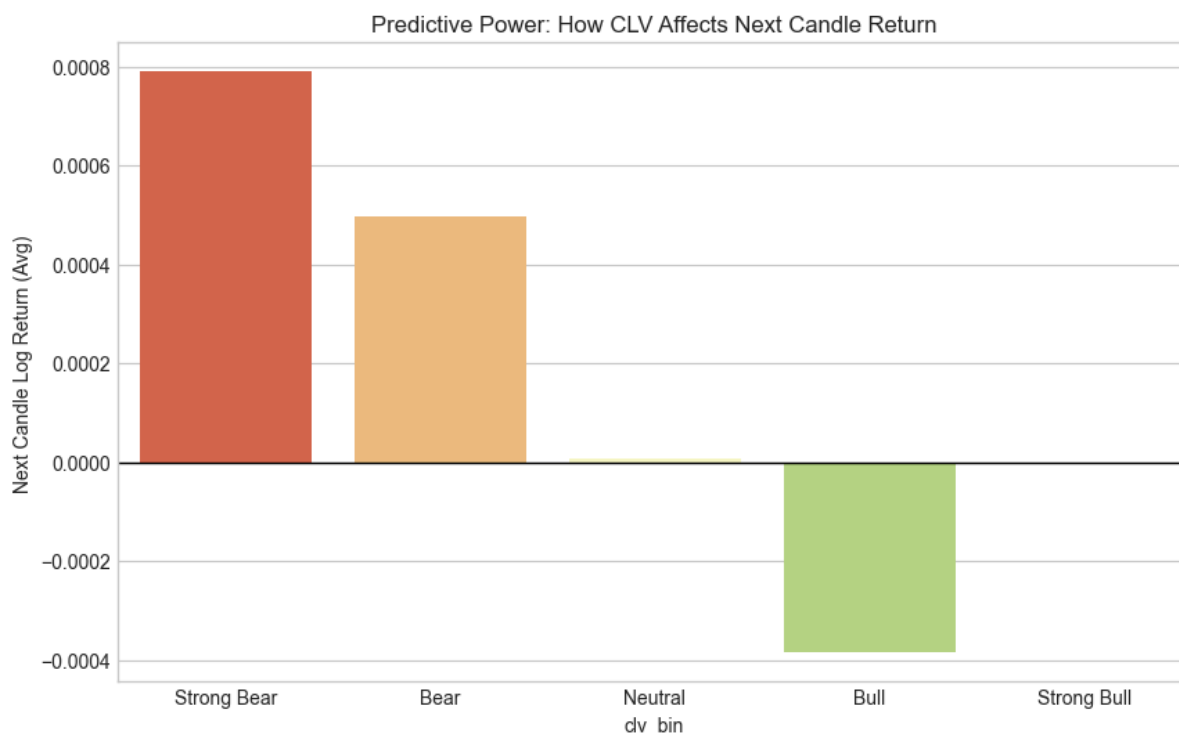


Figure 2.4.8. Next-period returns across CLV buckets

4.5.2 Wick imbalance

Beyond the close location, wick length provides information about rejection and absorption. Wick imbalance measures asymmetry between the upper wick and lower wick, indicating whether price was rejected at higher levels (supply pressure) or lower levels (demand pressure). This is a common price action microstructure signal and complements CLV.

4.5.3 Retained features

Two microstructure features are retained. The close location value (CLV) is kept because it summarizes where the candle closes within its range, providing a compact measure of late-session dominance by buyers or sellers. Wick imbalance is retained as a proxy for rejection and absorption dynamics, capturing whether price was rejected more strongly at the high or the low. Together, these features encode behavioral information that is not captured by returns or volatility alone.

4.6 Technical indicators analysis and feature engineering (bollinger bands)

4.6.1 Objective and financial intuition

In price data, many signals are not contained in absolute price levels but in the relative position of price versus its typical state. Bollinger Bands are introduced to normalize price location within a dynamic band and to provide a direct indicator of volatility regime. In short-term return prediction, these indicators provide complementary context to microstructure and volatility: instead of intrabar behavior, Bollinger Bands capture short-term “context” through rolling statistics.

4.6.2 Ex-ante design and future leakage prevention

Rolling statistics (mean and standard deviation) are computed over a 20-period window and shifted one step backward. This ex-ante design ensures that at time t , contextual statistics are formed using information observed up to $t-1$, avoiding unintended leakage from the future.

4.6.3 %B and width: normalized position and volatility regime

Two features are extracted from Bollinger Bands:

- (1) Bollinger %B normalizes the close price position within the band, enabling consistent interpretation across high- and low-price regimes. %B is not strictly bounded within $[0, 1]$; values outside the band indicate breakouts, which may correspond to strong momentum or overbought/oversold conditions.
- (2) Bollinger width normalizes band width by the moving average level, reflecting volatility regimes. Rising width is typically associated with volatility expansion, while low width reflects a “squeeze,” which often precedes breakouts.

Visual inspection shows that %B stays near the center during sideways phases but forms clusters outside the band during volatile breakouts. Bollinger width exhibits clear expansion–contraction cycles consistent with volatility clustering.

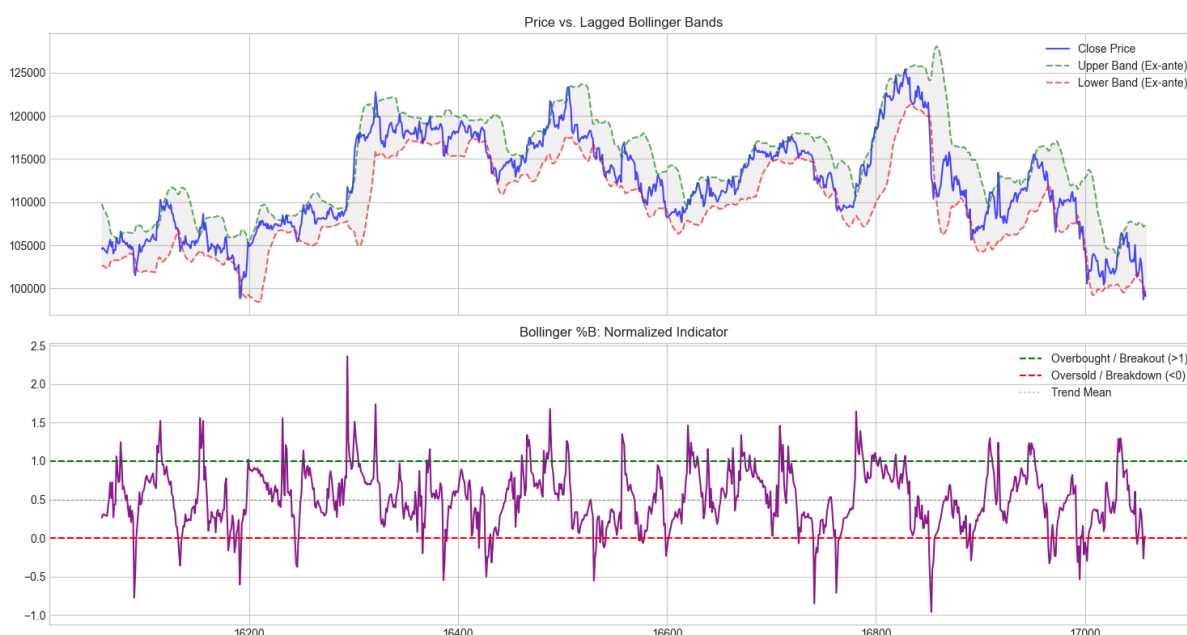


Figure 2.4.9. Bollinger %B and Bollinger width over time, capturing relative price position and volatility regimes

4.6.4 Retained features

Two Bollinger-derived features are retained. Normalized band position (Bollinger %B) is kept because it provides a scale-invariant interpretation of where price sits

relative to its short-term distribution, including breakout-like conditions. Band width is retained as a direct proxy for the volatility regime, capturing squeeze and expansion phases that frequently precede large moves. Both are constructed using an ex-ante rolling design to avoid future leakage.

4.7 Inter-market correlation analysis and feature engineering

Crypto markets are highly integrated—BTC often leads, while ETH and other assets tend to follow. Measuring BTC–ETH correlation therefore provides broader inter-market context.

4.7.1 Objective and economic intuition

BTC often drives market-wide movements, while ETH can move in sync or decouple depending on regime (e.g., network upgrades, narratives, or expectation shifts). Therefore, the correlation between BTC and ETH log returns provides information about market synchronization, enabling the model to incorporate inter-market context rather than relying on BTC-only signals.

4.7.2 Window design and regime interpretation

Correlation is computed using a 30-day rolling window, equivalent to 180 H4 periods, balancing stability and responsiveness to regime changes. The visualization suggests:

- (1) Many periods exhibit high correlation, reflecting tightly integrated markets with BTC as a dominant driver.
- (2) Correlation drops tend to coincide with prolonged consolidation or ETH-specific phases where ETH temporarily decouples.

These regime changes provide important context for prediction: the same BTC price action signal may have different implications under high-correlation regimes versus decoupling regimes.

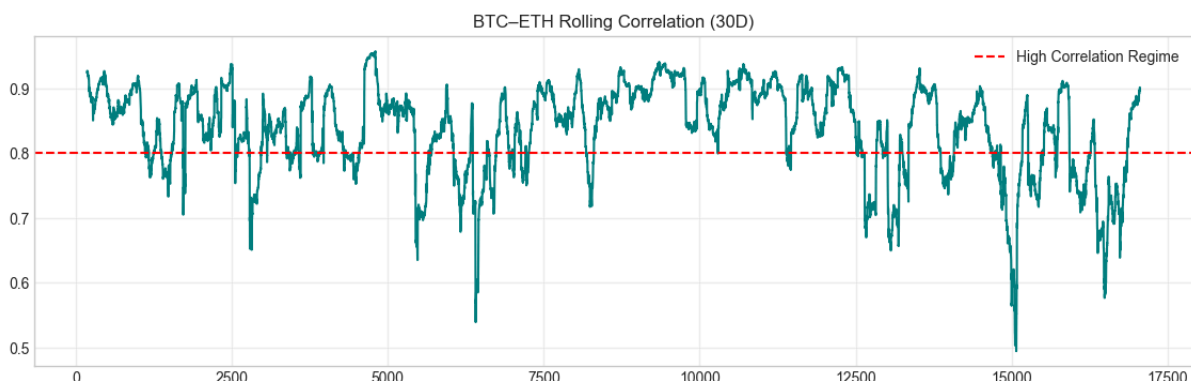


Figure 2.4.10. Rolling BTC–ETH correlation (30 days)

4.7.3 Retained features

The rolling bitcoin–ethereum return correlation is retained to represent inter-market context. It quantifies whether the market is in a highly synchronized regime or in a decoupling regime, which can change the meaning of otherwise similar BTC signals. Using a sufficiently long rolling window balances stability and responsiveness to regime changes.

4.8 Quote volume and trades analysis and feature engineering

4.8.1 Average ticket size (ATS): proxy for average order size

In trading data, quote volume reflects traded notional value, while the number of trades reflects order flow fragmentation. Their ratio, ATS, serves as a proxy for average trade size: high ATS often indicates larger participants (whales/institutions), while low ATS suggests retail-dominated activity.

4.8.2 Log relative ATS: ex-ante normalized ATS

Raw ATS tends to increase over time due to price level changes and market expansion. Therefore, ATS is normalized relative to its 20-period moving average using an ex-ante shift, and then log-transformed to produce a more stable series centered around zero. Visual evidence shows that raw ATS exhibits a long-term upward trend, whereas the normalized ATS reacts more clearly to abnormal periods (e.g., clusters of large trades).

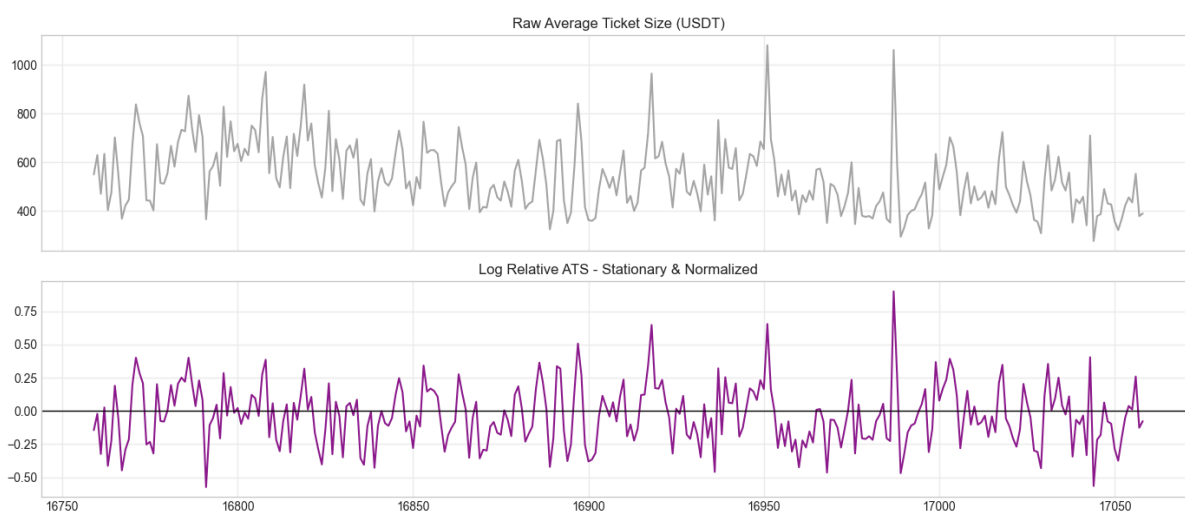


Figure 2.4.11. Raw ATS and log relative ATS

4.8.3 Feature selection decision

A single feature is retained: relative average trade size (log-normalized average ticket size). It is kept because it provides a proxy for participant composition—periods dominated by larger trades may indicate stronger institutional or whale activity, which can affect short-term price dynamics. In contrast, trade-count momentum is excluded

because its empirical association with the target is weak and it adds dimensionality without meaningful incremental information.

4.9 Market capitalization analysis and feature engineering

4.9.1 Overview and economic rationale

Market capitalization of BTC, ETH, and stablecoins provides insight into market-wide capital structure. Log-scale visualization typically shows BTC and ETH at similar orders of magnitude, while USDT and USDC are smaller but act as key liquidity instruments in the ecosystem. In crypto markets, stablecoins are often treated as dry powder: increases in stablecoin supply may reflect new capital inflows (fiat on-ramp) and rising buying power for risk assets.

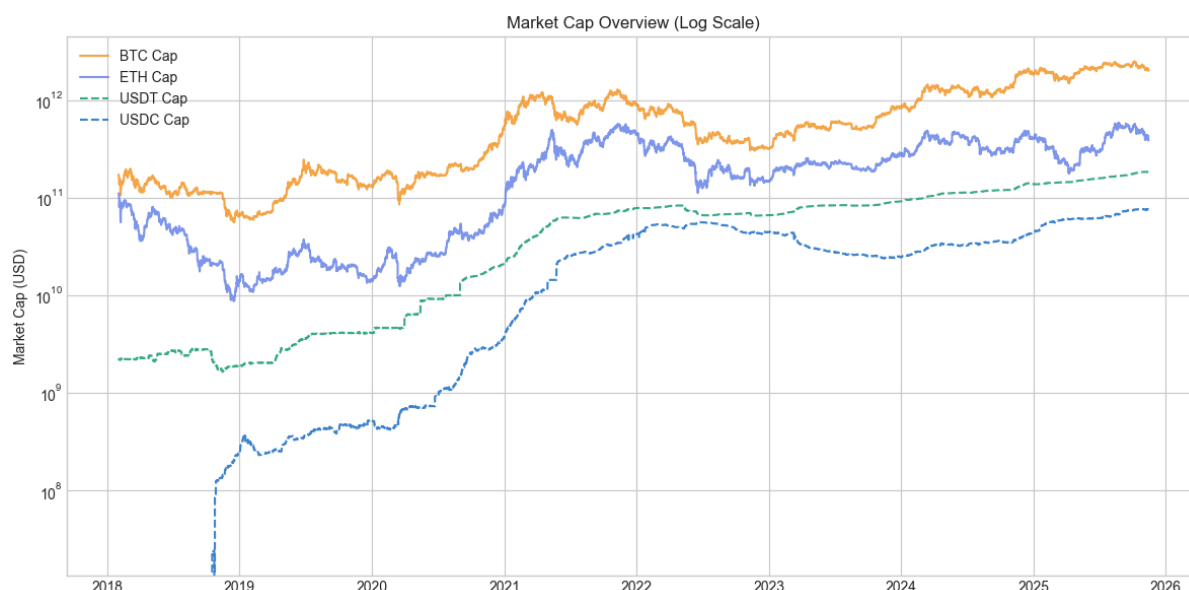


Figure 2.4.12. Log-scale market capitalization of BTC, ETH, USDT, and USDC, illustrating size structure and the role of stablecoins

4.9.2 Stablecoin supply change: stablecoin liquidity flow

Stablecoin supply change is constructed from total stablecoin market cap (USDT + USDC) to reflect expansion or contraction in stablecoin liquidity. The signal fluctuates around zero but exhibits spikes during strong minting or redemption phases, consistent with liquidity shock intuition.

4.9.3 SSR z-score: regime-based buying power

SSR measures the relationship between BTC scale and available stablecoin dry powder. A high SSR suggests relatively low buying power (less stablecoin relative to BTC cap), indicating potentially “heated” conditions; a low SSR suggests higher buying power and potentially more room for upside. Standardizing SSR using a rolling z-score makes extremes and mean-reversion behavior more apparent.

4.9.4 USDT dominance change: a risk-on/risk-off proxy

Changes in USDT share within the local market cap aggregate (BTC + ETH + stablecoins) reflect shifts from risk assets to stablecoins or vice versa. Empirically, USDT dominance change often behaves counter-cyclically relative to price action, making it a useful explanatory variable for risk-off regimes.

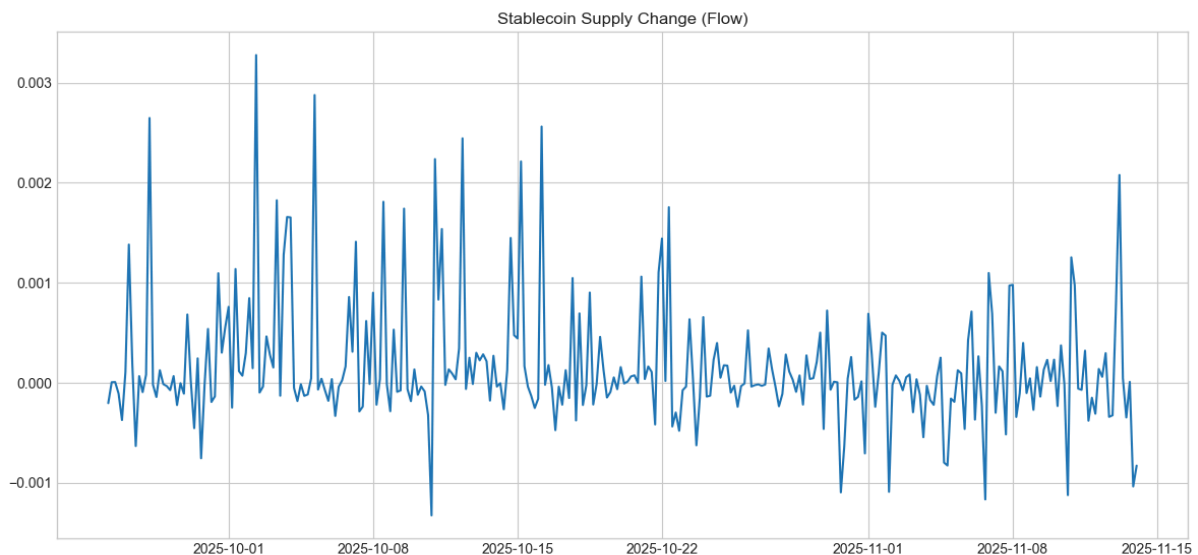


Figure 2.4.13a. Stablecoin supply change

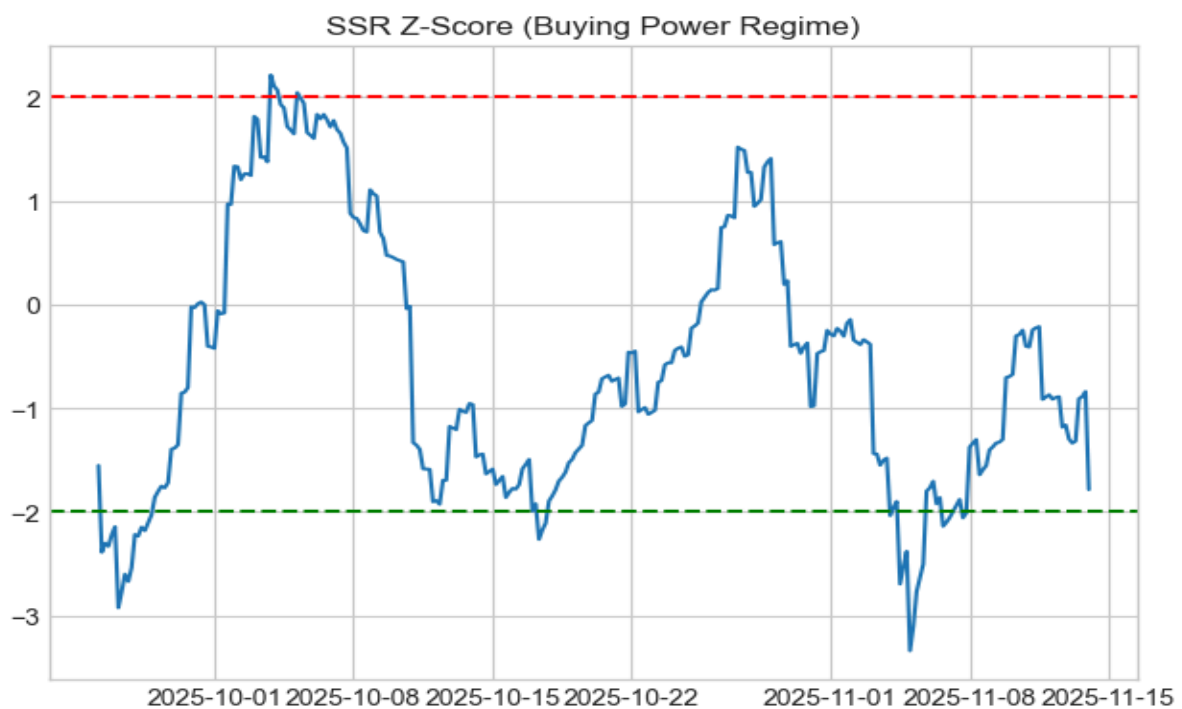


Figure 2.4.13b. SSR z-score and USDT dominance change

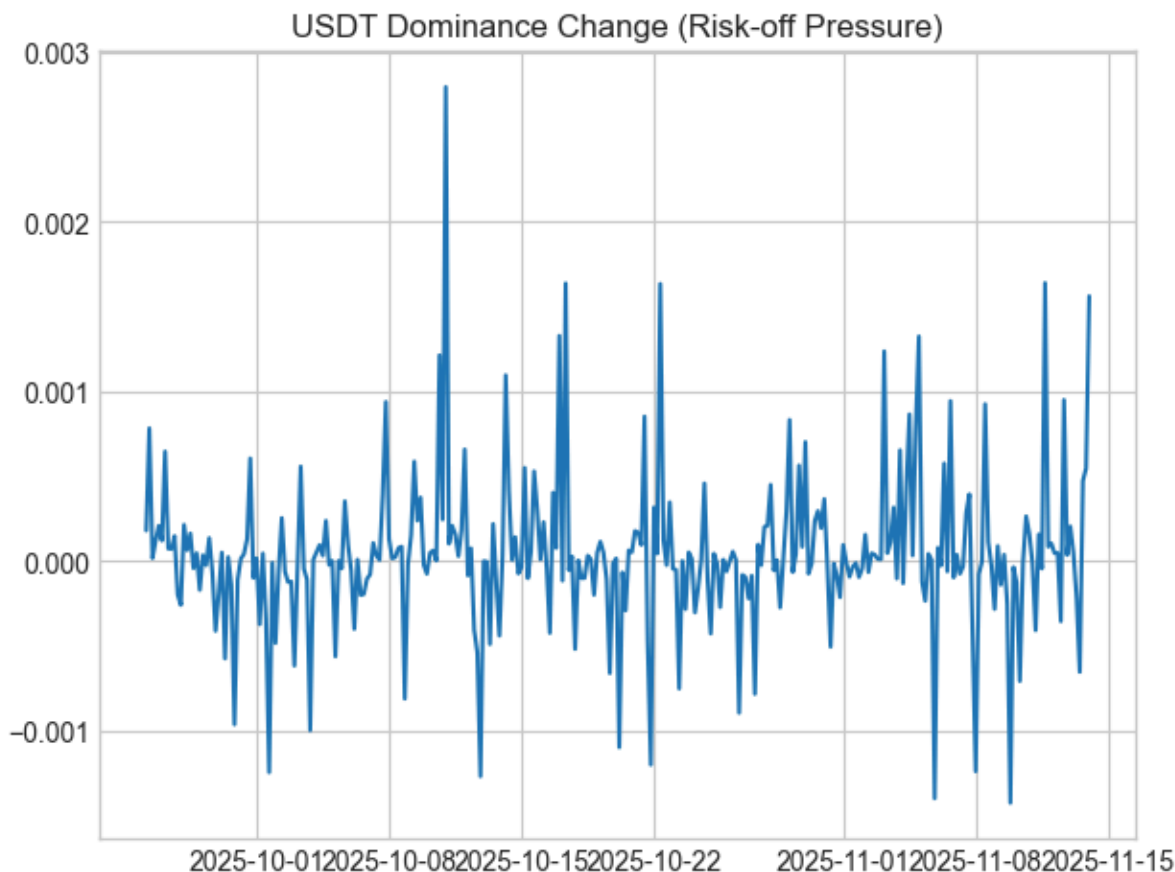


Figure 2.4.13c. Liquidity and risk-on/risk-off signals

4.9.5 Retained features

Three market-structure features are retained. Stablecoin supply change is kept as a liquidity flow indicator that proxies “dry powder” entering or leaving the ecosystem. Regime-normalized stablecoin-to-bitcoin buying power (SSR z-score) is retained to capture valuation and liquidity extremes relative to historical conditions. Finally, USDT dominance change is included as a risk-on/risk-off proxy, reflecting shifts between stablecoin preference and risk-asset exposure. These variables complement price-derived signals by representing capital availability and allocation.

4.10 Network activity analysis and feature engineering

4.10.1 Address growth: adoption momentum

Active addresses serve as a direct proxy for network participation. Rather than using absolute levels, the study uses log-based rate of change to reduce scale effects and improve stability, capturing adoption momentum.

4.10.2 Transaction intensity: transactions per user

Total transaction count may increase simply because the user base grows. Therefore, transaction intensity is measured as transactions per active address, and then differenced to capture changes in average user engagement.

4.10.3 Network valuation z-score: relative valuation versus usage

The ratio between price (or a market valuation proxy) and usage (active addresses) follows a logic similar to price-to-fundamental valuation in traditional finance. When standardized as a z-score using a 90-period rolling window, it exhibits mean-reversion properties and highlights valuation extremes relative to historical behavior.

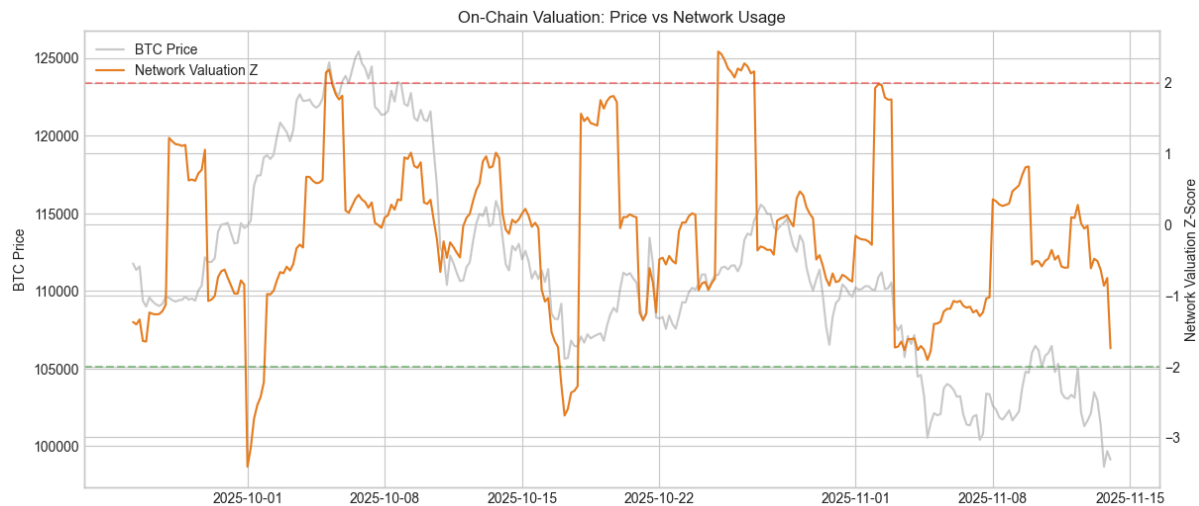


Figure 2.4.14. Network valuation z-score, where extremes indicate over/under-valuation relative to usage

4.10.4 Retained features

Three network activity features are retained. Active address growth (log-change) is kept as a stationary proxy for adoption momentum. Transaction intensity per active address (change) is retained to capture shifts in average user engagement beyond pure user growth. Regime-normalized network valuation relative to usage (z-score) is included to represent valuation extremes in a “price-to-usage” sense, which can inform mean-reversion or overheating conditions.

4.11 Mining analysis and feature engineering

4.11.1 Hash ribbon spread: miner capitulation and recovery

Hash rate reflects network security strength and miners’ economic incentives. The spread between short-term and long-term hash rate moving averages helps identify capitulation phases and recovery phases. Empirically, capitulation often coincides with prolonged market drawdowns, while recovery tends to align with improving market conditions.

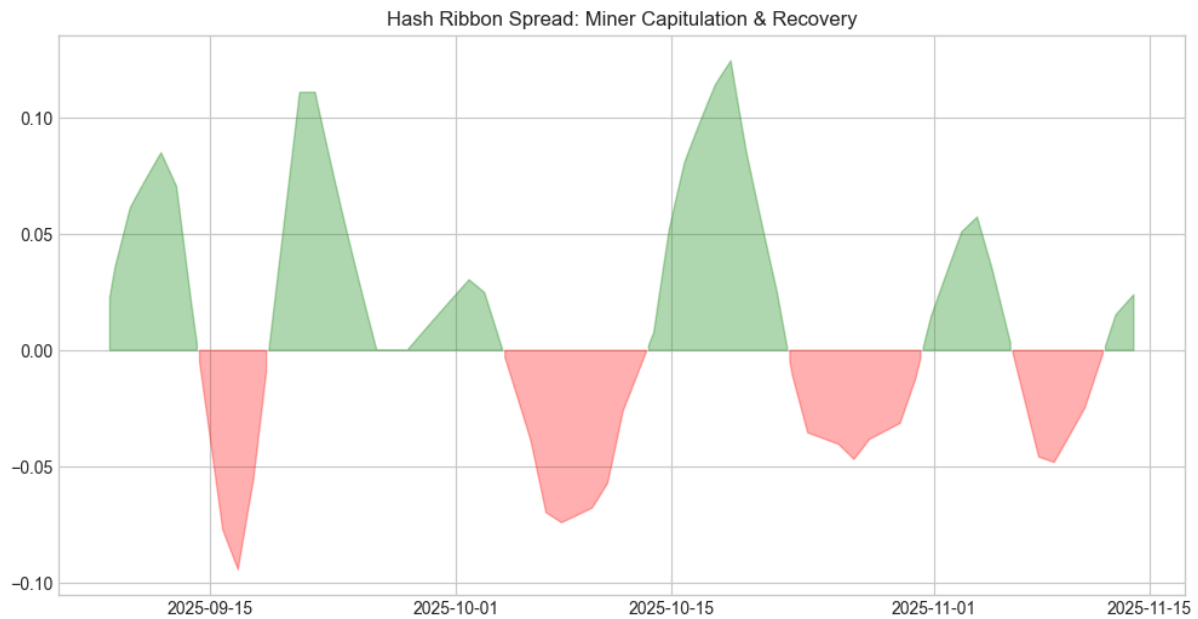


Figure 2.4.15a. Hash ribbon spread and log Puell multiple

4.11.2 Puell multiple: miner selling pressure

When miner revenue is above normal levels, the incentive to realize profits (selling pressure) increases; when revenue is low, selling pressure may decline. Normalizing by a long-term average (365) and applying a log transform yields a more stable variable suitable for modeling.



Figure 2.4.15b. Capitulation/recovery signals and miner selling pressure

4.11.3 Retained features

Two mining-related features are retained. The hash ribbon spread is kept because it captures miner capitulation and recovery dynamics by contrasting short- and long-term hash rate trends; these phases often align with broader market stress and subsequent normalization. The Puell multiple (log-transformed) is retained as a proxy for miner revenue pressure relative to its historical baseline, which can influence miners' propensity to sell and therefore affect supply pressure. Both features are used in normalized or transformed forms to improve stability and reduce scale effects.

4.12 On-chain metrics analysis and feature engineering

4.12.1 MVRV ratio z-score: market valuation versus realized value

MVRV combines price action with on-chain cost basis via realized capitalization, making it a powerful valuation signal in crypto. Standardizing with a long-window z-score allows it to reflect regime-based valuation extremes.

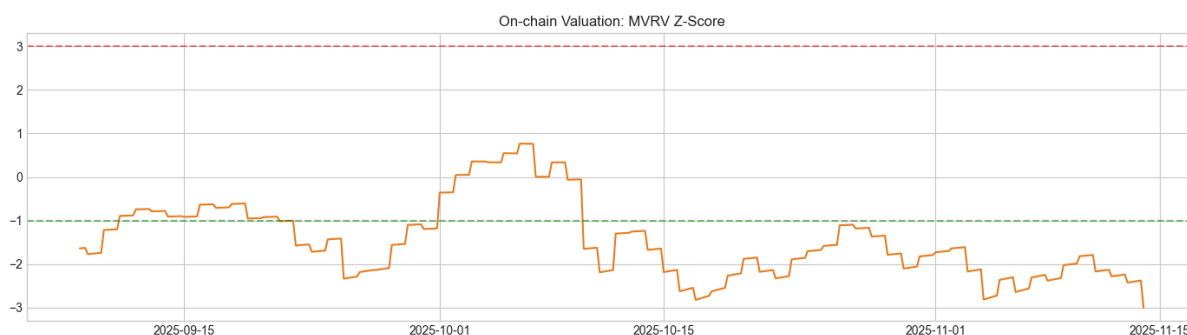


Figure 2.4.16a. MVRV z-score

4.12.2 Exchange netflow z-score: on-chain selling pressure

Net exchange inflow minus outflow proxies preparation to sell (inflow) versus accumulation or self-custody (outflow). Standardizing netflow with a z-score enables the model to detect flow shocks rather than relying on absolute levels.

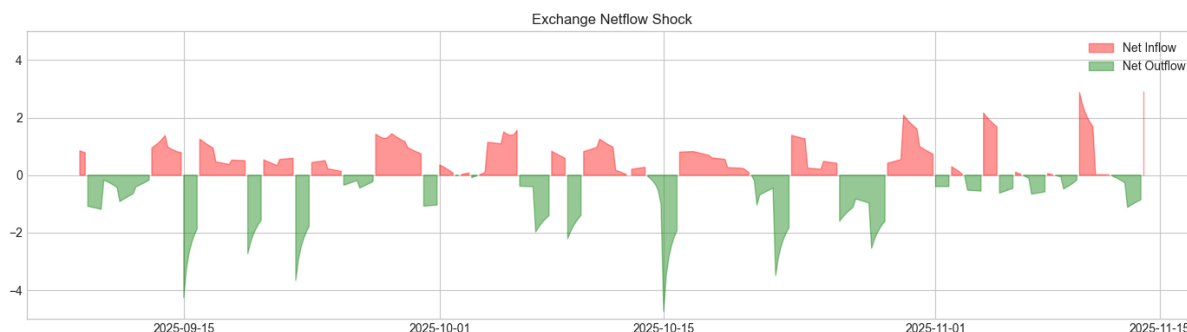


Figure 2.4.16b. Exchange netflow z-score

4.12.3 Exchange reserve ratio change: exchange supply structure

The share of supply held on exchanges reflects readily tradable supply. Changes in reserve ratio provide signals of deposit trends (increasing selling overhang) versus withdrawals (reducing selling overhang).

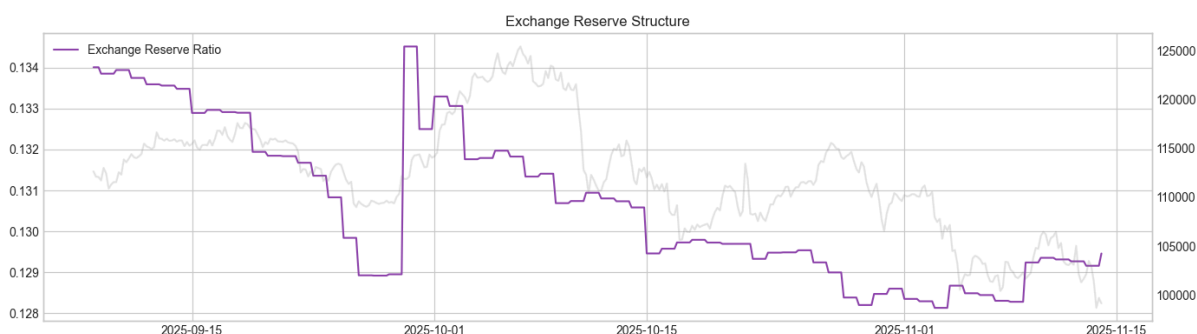


Figure 2.4.16c. Reserve ratio change

4.12.4 Retained features

Three on-chain features are retained. The MVRV ratio (z-scored) is kept as a regime-aware valuation indicator comparing market value to realized value, helping identify valuation extremes. The exchange netflow (z-scored) is retained to capture flow shocks that proxy potential selling or accumulation pressure. The exchange reserve ratio change is included to reflect structural shifts in readily tradable supply on exchanges, providing a supply-side context that price-only features cannot capture.

4.13 News analysis and feature engineering

4.13.1 Overview of the news data

News data are sparse over time: many periods contain no articles, interspersed with bursty clusters around major events. This makes it important to distinguish between “no news” and “news with neutral sentiment.” Visualizing news intensity reveals spikes that coincide with high-volatility market phases, consistent with the intuition that news acts as a catalyst for price movements.



Figure 2.4.17. News frequency over time

4.13.2 News impact score: attention-weighted impact

Mean sentiment captures direction (bullish/bearish) but ignores attention intensity. Therefore, the impact score combines mean sentiment with a log function of article count, reflecting that marginal impact does not scale linearly with the number of articles. This design helps differentiate between a period with a single article and a period with multiple aligned articles.

4.13.3 News sentiment decay (slow): time-decayed effect

News often does not impact the market within a single candle; effects can persist and gradually decay. The study therefore applies an exponentially weighted moving (EWM) decay signal to smooth noise from individual shocks and to capture the “impact tail” of information.

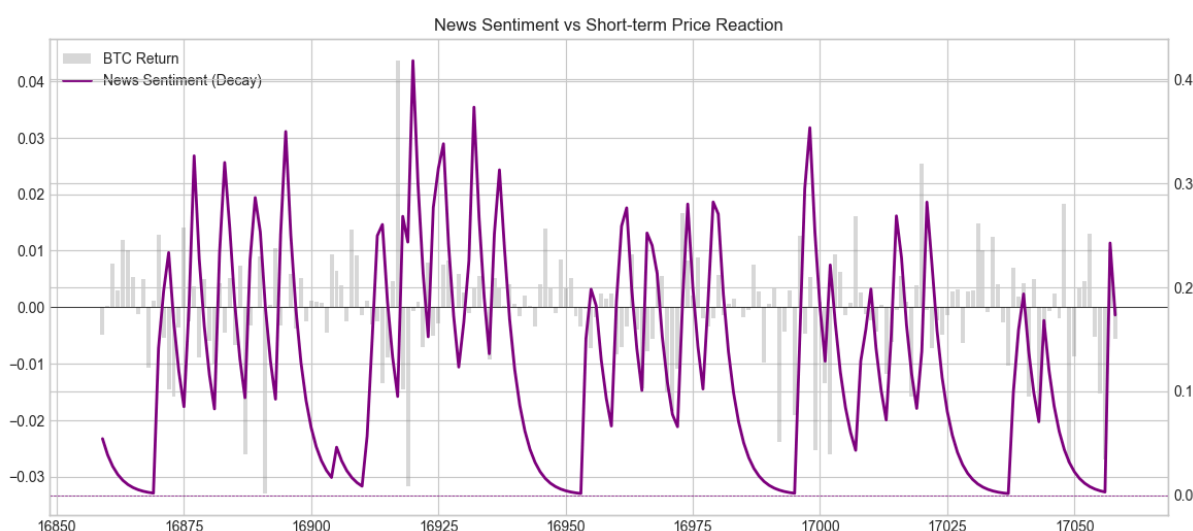


Figure 2.4.18. News sentiment impact in short-term price dynamics

The slow decay (purple line) smooths noise from individual news shocks and tracks short-term price reactions, suggesting that news impacts price almost immediately.

4.13.4 News sentiment shock: abrupt changes

In addition to the impact level, a shock feature is constructed to measure abrupt changes between consecutive periods. High positive shocks often correspond to surprise events, while strong negative shocks correspond to negative surprises.

4.13.5 Retained features

Four news-derived features are retained to address both direction and intensity. Article count is kept to distinguish “no news” from active information periods. The attention-weighted impact score is retained to incorporate sentiment direction while accounting for diminishing marginal impact of additional articles. The slow decay sentiment signal is included to represent persistent information effects beyond a single

candle. Finally, the sentiment shock feature is retained to detect abrupt changes consistent with surprise events.

4.14 Sentiment index analysis and feature engineering (Fear & Greed)

4.14.1 Overview of the Fear & Greed Index

The Fear & Greed Index is a composite measure on a 0–100 scale and is commonly categorized into five levels from extreme fear to extreme greed. In the sample, the index varies widely, reflecting cyclical shifts in market psychology. Instead of using only the level, two transformations are prioritized: momentum and divergence from the moving average, consistent with behavioral finance arguments that extremes are often associated with turning points.

4.14.2 FNG change: sentiment momentum

FNG change measures period-to-period change, capturing the speed at which sentiment improves or deteriorates. Empirically, large changes often coincide with regime transitions, as crypto market sentiment can flip rapidly in response to news or sharp price moves.

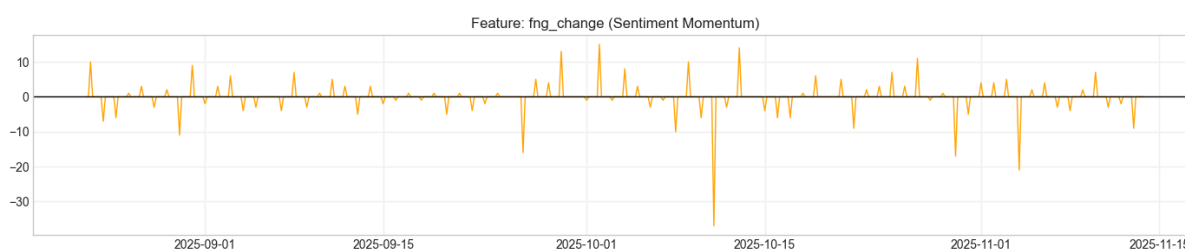


Figure 2.4.19. FNG change over time

4.14.3 FNG divergence: contrarian signal versus MA7

FNG divergence measures the gap between the current level and the 7-day moving average (equivalent to 42 H4 periods). Positive divergence indicates greed above normal levels and may imply overheating, while negative divergence indicates elevated fear. Bucket-based analysis shows that the distribution of absolute returns differs across extreme fear and extreme greed regimes, suggesting that sentiment regimes are associated with distinct risk conditions.

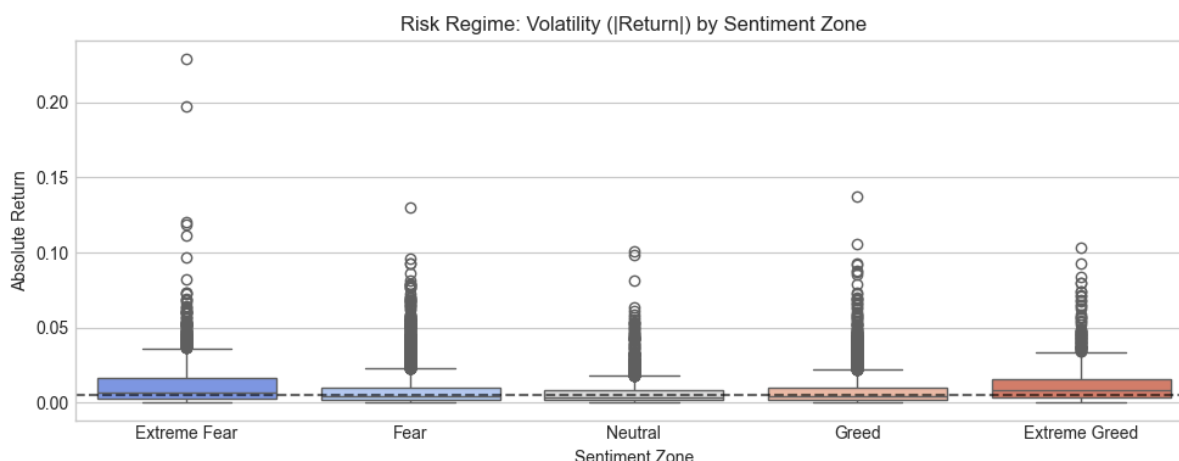


Figure 2.4.20. Absolute return across sentiment buckets

4.14.4 Retained features

Two sentiment-index transformations are retained. Sentiment momentum (change) is kept because rapid shifts in market psychology often coincide with regime transitions. Sentiment divergence from a short-term baseline is retained as a contrarian-style indicator that highlights extreme deviations from typical sentiment, which can be informative under behavioral finance dynamics.

4.15 Feature engineering summary

4.15.1 Correlation with the target and weak-signal characteristics

Linear correlation between each feature and future returns is computed to assess direct association. Results show that individual correlations are small, consistent with the weak-signal nature of financial return prediction: markets are noisy and signals are typically weak at the single-variable level. However, several feature groups (on-chain valuation, exchange flows, microstructure, and certain momentum/regime features) exhibit relatively stronger correlations than the overall baseline. This supports the view that machine learning models can add value by combining multiple weak signals into a more informative forecast.

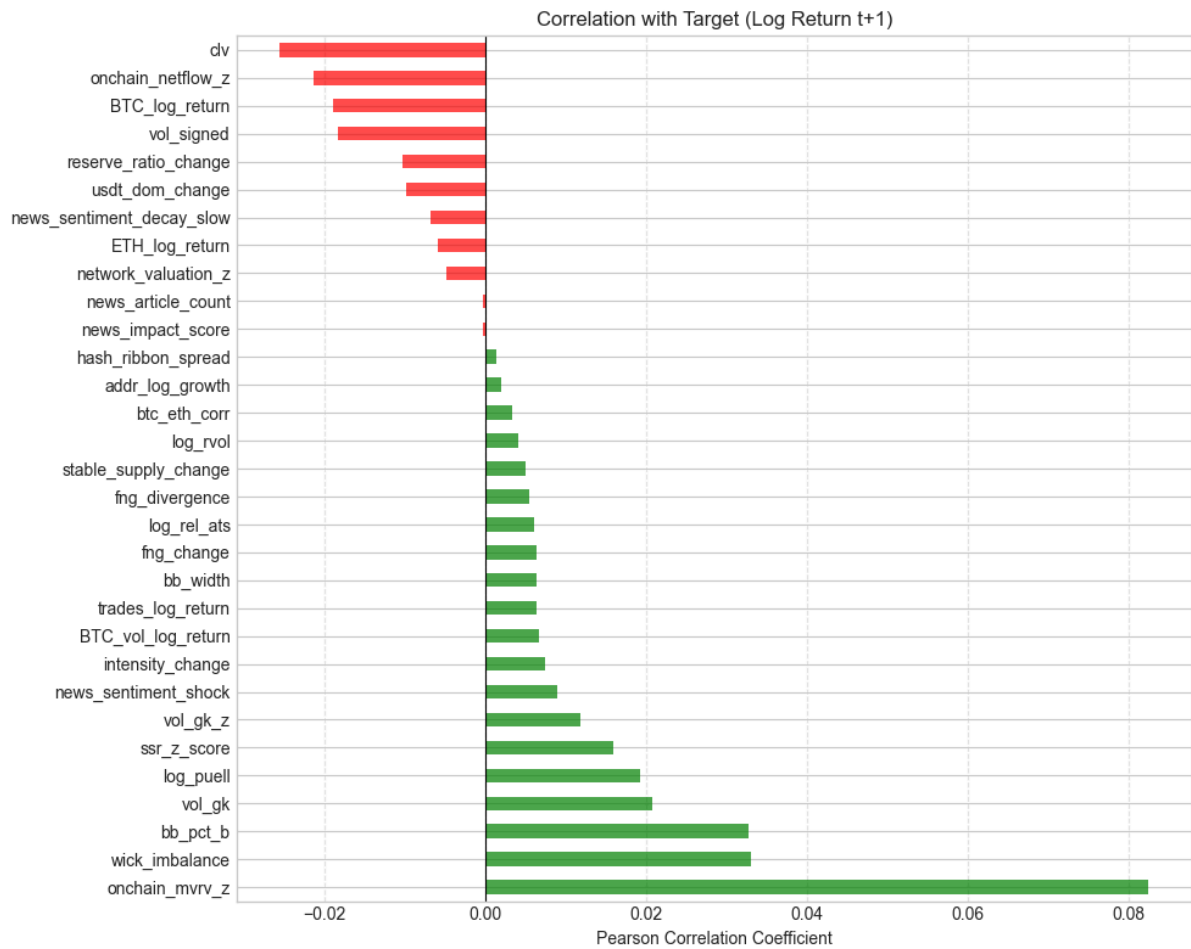


Figure 2.4.21. Top features by correlation with the target

4.15.2 Multicollinearity check

The correlation matrix indicates that no pair of features exceeds a high-correlation threshold (e.g., 0.85). Some moderate correlation clusters are expected (e.g., volatility and volatility z-score, or valuation-related indicators), but they do not pose severe multicollinearity risk. This confirms that the feature set is suitable for both linear models (ridge/lasso) and nonlinear models.

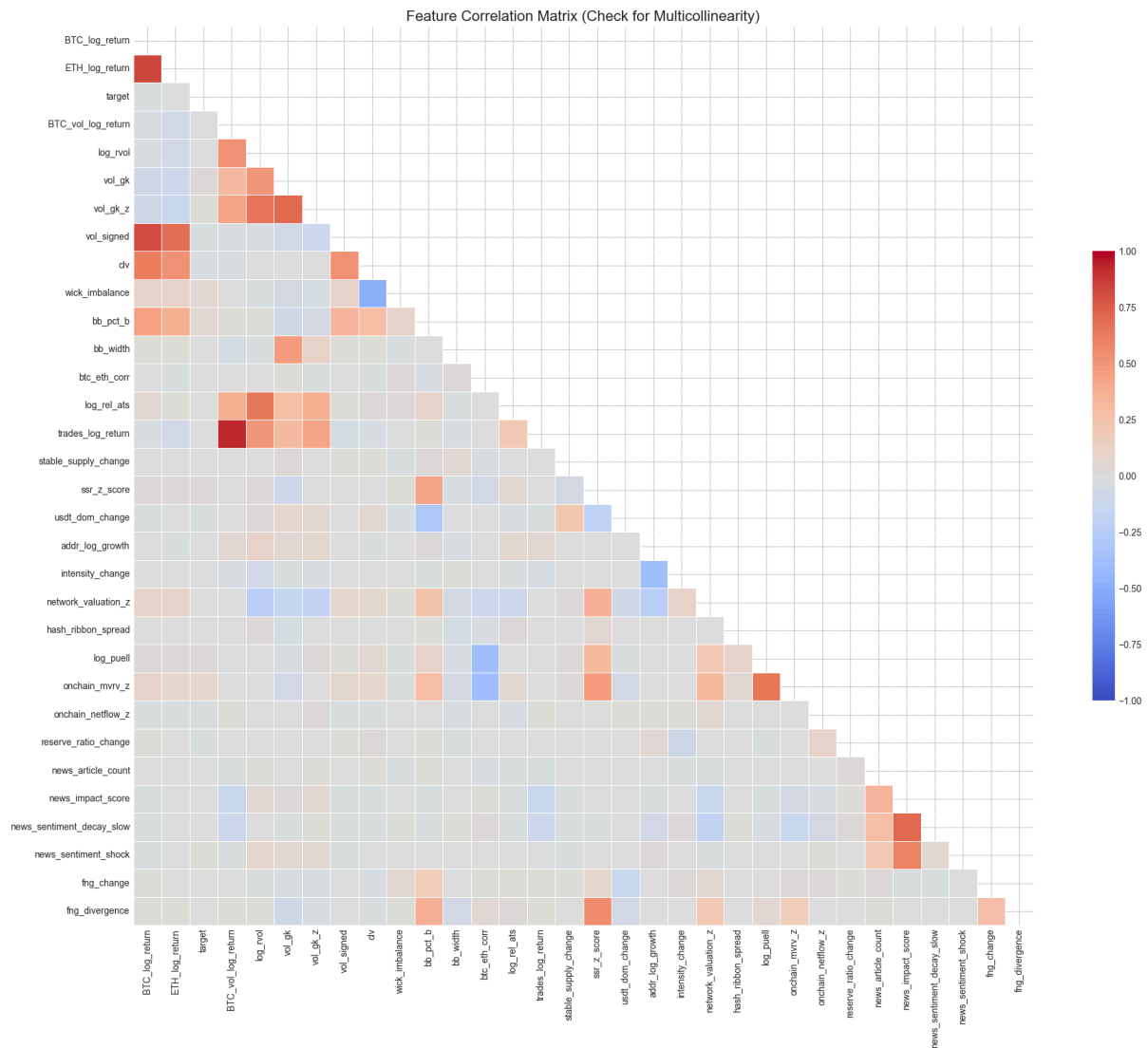


Figure 2.4.22. Feature correlation matrix

4.15.3 Rolling-window warm-up and the train-ready dataset

Because many features use rolling windows of varying lengths, early observations lack sufficient history and produce NaN values. The study addresses this by removing all rows containing NaNs after feature engineering. The resulting train-ready dataset has the following size:

Before warm-up removal: 17,059 observations

After warm-up removal: 16,328 observations

Retention rate: 95.7%

Final shape: 32 columns (31 predictors + 1 target)

After this step, the dataset contains no infinite values and is ready for model training.

4.16 Chapter conclusion

This chapter presented the EDA and feature engineering pipeline that transforms 60 raw variables from seven heterogeneous data sources into a stationary, economically meaningful feature set suitable for 4-hour bitcoin return prediction. Core transformations include log transforms and differencing to address non-stationarity, regime normalization via z-scores to detect extremes and flow shocks, and behavioral features from microstructure and technical indicators. To ensure deployment feasibility, all rolling statistics follow an ex-ante design to minimize future leakage. After removing the rolling-window warm-up period, the train-ready dataset retains 95.7% of observations and provides 31 predictors, serving as direct inputs for Chapter 3 on model training and evaluation.

CHAPTER III. MODEL DEVELOPMENT AND EVALUATION

1. Experimental design

1.1 Problem formulation and data splitting strategy

The objective of this chapter is to train, evaluate, and select the optimal machine learning model for predicting bitcoin returns on the 4-hour timeframe. This is a regression task on financial time-series data, characterized by a low signal-to-noise ratio and weak-form market efficiency.

To realistically reflect a trading setting, the data splitting strategy follows a chronological split rather than random sampling. This approach prevents future information leakage and ensures that the training data do not contain information from the validation or test sets. The train-ready dataset consists of 16,328 samples and is divided into three subsets:

Training set (train): 70% (11,429 samples, from 2018-06-02 to 2022-10-17), used to optimize model parameters.

Validation set (validation): 15% (2,449 samples, from 2022-10-17 to 2024-03-20), used for hyperparameter tuning and monitoring overfitting during training.

Test set (test): 15% (2,450 samples, from 2024-03-20 to 2025-11-13), used as unseen data to evaluate the model's generalization performance.

1.2 Sliding-window feature design

To capture temporal dependencies and momentum/reversal patterns in time series, this study uses a sliding window with a window size of 6, corresponding to 24 hours of historical information (6 candles $\times$ 4 hours per candle). Each input sample contains 180 features (31 features $\times$ 6 lags), enabling the model to observe recent dynamics across four key dimensions: price action, market structure, on-chain health, and market sentiment.

1.3 Model selection

To identify the most suitable model, the study evaluates ten machine learning models across four algorithm families: linear regularized models, robust linear models, support vector machines (SVM), and tree-based ensembles. The evaluated models include Ridge, Lasso, ElasticNet, Huber, SGDRegressor, LinearSVR, SVR with an RBF kernel, Decision Tree, Random Forest, and Gradient Boosting.

In addition, two baseline models are used for comparison: Zero baseline (always predicts return = 0) and Last baseline (predicts the next return as the previous candle's return, representing a naive momentum strategy).

2. Evaluation metrics

2.1 Statistical metrics

To evaluate predictive quality, the study uses four primary statistical metrics. Root mean squared error (RMSE) measures average prediction error and is sensitive to outliers. Mean absolute error (MAE) provides a more robust measure that is less affected by extreme values. The Spearman information coefficient (IC) measures rank correlation between predictions and realized returns and is widely used in quantitative finance because it focuses on ranking ability rather than absolute accuracy. Finally, directional accuracy measures the proportion of cases in which the model correctly predicts the direction of return (up vs. down), providing a direct assessment of trend prediction capability.

2.2 Trading performance metrics

To assess practical value, the study simulates a trading strategy with an initial capital of USD 1 billion. The strategy follows basic trading rules: open a long position when the predicted return is positive, and close the position when the predicted return is negative or neutral. Based on this simulation, four financial metrics are computed.

Total PnL measures cumulative profit and loss across all trades. Win rate measures the proportion of trades with positive returns. Maximum drawdown (MDD) measures the largest peak-to-trough decline in equity, reflecting tail risk exposure. The Sharpe ratio is computed as the ratio between the mean and the standard deviation of per-trade returns, providing a risk-adjusted performance measure.

3. Training and evaluation results

3.1 Overall model performance

Table 3.1 summarizes the performance of the ten evaluated models together with the two baselines, sorted by Spearman IC on the test set (unseen data). The table compares four key metrics: Spearman IC (ranking ability), RMSE (prediction error), directional accuracy, and PnL (financial return).

Lasso achieves the best overall performance with $IC = 0.269$, $RMSE = 0.00937$, directional accuracy = 57.86%, and $PnL = +5.28$ billion USD. It is followed by Huber ($IC = 0.264$, $PnL = +5.10$ billion USD) and Ridge ($IC = 0.245$, $PnL = +4.90$ billion USD). The top four models are all linear regularized methods, each achieving $IC > 0.24$ and $PnL > 4.7$ billion USD, indicating that regularized linear models are particularly effective in this setting.

In contrast, ensemble models (Random Forest, Gradient Boosting) achieve IC values around 0.09 and PnL around 2 billion USD. The Last baseline incurs losses with $IC = -0.050$ and $PnL = -0.54$ billion USD, suggesting that naive momentum is not effective for this task.

Ranking	Model	Spearman IC	RMSE	Dir. Accuracy	PnL (USD bn)	Win Rate
1	Lasso	0.269	0.00937	57.86%	+5.28	57.86%
2	Huber	0.264	0.01574	58.76%	+5.10	58.76%
3	Ridge	0.245	0.00913	57.45%	+4.90	57.45%
4	ElasticNet	0.246	0.01584	56.26%	+4.77	56.26%
5	LinearSVR	0.148	0.03707	54.05%	+3.36	54.05%
6	SVR_RBF	0.141	0.01139	53.15%	+2.80	53.15%
7	GradientBoosting	0.097	0.00966	52.74%	+1.99	52.74%
8	RandomForest	0.090	0.00953	51.80%	+2.48	51.80%
9	DecisionTree	0.042	0.00969	50.82%	+0.83	50.82%
10	SGDRegressor	0.014	-	51.29%	+0.12	51.29%
-	Last Baseline	-0.050	0.01388	46.73%	-0.54	46.73%

Table 3.1. Model results on the validation/test evaluation (sorted by Spearman IC).

Lasso improves by 638% over the baseline in terms of IC (from -0.050 to 0.269) and by 1,078% in terms of PnL (+5.28 vs. -0.54 billion USD). The fact that the top four models are all linear regularized methods suggests that the problem contains a strong linear component.

3.2 Generalization performance

Table 3.2 evaluates generalization for the top three models by comparing performance on the validation set (2022–2024) and the test set (2024–2025). The “Change (%)” column indicates degradation or improvement when moving to future data.

Model	Metric	Validation	Test	Change (%)	Assessment
Lasso	Spearman IC	0.299	0.269	-10.0%	normal degradation
	RMSE	0.00945	0.00937	-0.85%	improved
	Dir. Accuracy	57.84%	57.86%	+0.02%	fully stable
	PnL (tỷ USD)	+5.69	+5.28	-7.2%	slight degradation
	Max Drawdown	-0.11	-0.09	-18.2%	lower risk
Huber	Spearman IC	0.295	0.264	-10.5%	slight degradation
	PnL (tỷ USD)	+5.69	+5.10	-10.4%	moderate degradation
	Max Drawdown	-0.14	-0.13	-7.1%	stable
Ridge	Spearman IC	0.283	0.245	-13.4%	slight degradation
	PnL (tỷ USD)	+5.36	+4.90	-8.6%	moderate degradation
	Max Drawdown	-0.14	-0.08	-42.9%	lower risk

Table 3.2. Generalization performance: validation vs. test (top 3 models).

A mild reduction (approximately 7–10%) in IC and PnL from validation to test is expected due to market regime changes in the future period. Directional accuracy remains essentially unchanged at 57.86%, while maximum drawdown improves, indicating that the model does not overfit and generalizes well.

3.3. Comparison by algorithm family

Table 3.3 summarizes the best-performing model from each algorithm family, showing that linear regularized methods outperform alternative approaches.

Algorithm family	Best model	IC (Test)	RMSE (Test)	PnL (Test)	Win Rate	Key reason
Linear Regularized	Lasso	0.269	0.00937	+5.28	57.86%	effective regularization; strong features
Robust Linear	Huber	0.264	0.01574	+5.10	58.76%	robust to outliers; stable generalization
SVM	LinearSVR	0.148	0.03707	+3.36	54.05%	RBF tends to overfit; linear kernel underperforms
Tree-based	GradientBoosting	0.097	0.00966	+1.99	52.74%	overfits noise; high variance
Baseline	Last	-0.050	0.01388	-0.54	46.73%	naive momentum is ineffective

Table 3.3. Performance comparison by algorithm family (best model in each group).

Regularized linear models outperform primarily because: (1) the task has a strong linear component—signals are weak and embedded in noise, making nonlinear models more prone to overfitting; (2) features are carefully engineered—stationary, standardized, and economically interpretable; (3) L1/L2 regularization reduces overfitting and performs implicit feature selection; and (4) ensemble methods exhibit higher variance and tend to overfit in noisy financial data.

4. Analysis of the best model (Lasso regression)

4.1 Performance overview

Lasso (least absolute shrinkage and selection operator) is a linear regression model with L1 regularization, which delivers strong predictive performance while performing automatic feature selection. On the test set, the model achieves RMSE = 0.00937 (corresponding to a 0.937% error in log return), Spearman IC = 0.269

(positive rank correlation with future returns), directional accuracy = 57.86% (7.86 percentage points above random), total PnL = +5.28 billion USD (+528% on an initial capital of USD 1 billion), win rate = 57.86%, maximum drawdown = −9%, and Sharpe ratio = 0.206.

These results are practically meaningful. An IC of 0.269 indicates that the model can rank return opportunities and distinguish high-return from low-return periods. A directional accuracy of 57.86% is substantially above the 50% random baseline, suggesting genuine predictive power. The PnL-to-MDD ratio equals $5.28 / 0.09 = 58.7$, indicating an excellent risk–return trade-off, outperforming traditional buy-and-hold approaches.

4.2 Feature importance

Lasso produces a sparse weight vector, where some features are assigned zero coefficients and are effectively removed. Among the 180 input features (31 features × 6 lags), approximately 147 have non-zero coefficients, implying that about 18% of features are eliminated. This helps reduce overfitting and improves interpretability.

Rank	Feature	Absolute weight	Group	Economic interpretation	Most common lag
1	BTC_log_return	0.0523	Price	Short-term momentum	lag1 (42.5%)
2	MVRV_z_score	0.0487	On-chain	Valuation signal	lag1
3	Volatility_GK	-0.0412	Volatility	Mean reversion (high vol → lower return)	lag1
4	CLV	-0.0398	Microstructure	Reversal (close near high → lower return)	lag1
5	Bollinger_%B	0.0365	Technical	Momentum continuation	lag1
6	FNG_divergence	-0.0341	Sentiment	Contrarian (greed → correction)	lag2 (21.7%)
7	News_sentiment_decay	0.0328	News	Sentiment smoothed	lag1

8	ATS_log_relative	0.0305	Microstructure	Whale activity	lag3 (15.6%)
9	Vol_log_return	0.0294	Volume	Volume momentum	lag1
10	Hash_ribbon_spread	0.0287	Mining	Miner health (recovery signal)	lag2

Table 3.4. Top 10 features by absolute coefficient weight and lag distribution.

Lag distribution shows lag1 accounts for 42.5%, lag2 for 21.7%, and lag3 for 15.6%, totaling 79.7%, implying that the most recent 12 hours dominate the prediction signal. Lags 4–6 contribute only about 20% of total weight. The top features span price, on-chain, volatility, and microstructure dimensions, reflecting a combination of momentum, valuation, and market structure effects. Negative signs on volatility, CLV, and FNG divergence indicate mean-reversion behavior: high volatility, closes near the candle high, or excessive greed are typically followed by lower returns. The dominance of lag1 is consistent with short-horizon market dynamics, where the latest information is most informative.

4.3. Lasso vs. baselines

Table 3.5 compares Lasso against the two baseline models and reports relative improvements.

Metric	Zero Baseline	Last Baseline	Lasso	Improvement vs. last
IC Spearman	0.000	-0.050	0.269	+638%
RMSE	-	0.01388	0.00937	-32.5%
Directional Accuracy	50.00%	46.73%	57.86%	+23.8%
Win Rate	-	46.73%	57.86%	+23.8%
PnL (tỷ USD)	0.00	-0.54	+5.28	+1078%

Max Drawdown	0.00	-0.79	-0.09	+88.6% (8.8× lower risk)
Sharpe Ratio	0.000	-0.019	0.206	Positive

Table 3.5. Lasso vs. baselines: relative improvements (%).

The IC improvement of 638% shows that Lasso not only produces positive ranking power but does so substantially better than the baseline. PnL improves by 1,078%, turning a −0.54 billion USD loss into a +5.28 billion USD gain, an absolute swing of 5.82 billion USD. Risk is reduced by 8.8× as maximum drawdown decreases from −79% to −9%. Directional accuracy at 57.86% exceeds the 50% random baseline by a meaningful margin, suggesting the results are not driven by chance.

5. Limitations and practical considerations

Although the model performs well in simulation, several practical factors must be considered for deployment. First, the simulation does not include transaction fees (often around 0.1% per trade on Binance), which can materially reduce net returns. Second, slippage (the difference between expected and realized execution price) is not modeled, which becomes critical at large trading sizes (e.g., USD 1 billion). Third, the simulation assumes a fixed position size, whereas real trading systems often require dynamic position sizing based on the Kelly criterion or other risk management methods.

In addition, the model is only tested on bitcoin on the 4-hour timeframe. Performance may differ when applied to Ethereum, altcoins, or other time horizons. Finally, the model is static—parameters are trained once and not updated as new data arrive—introducing the risk of model drift under changing market regimes.

6. Chapter conclusion

This chapter has presented the training, evaluation, and model selection process for short-term (4-hour) bitcoin return prediction. The main findings show that regularized linear models, particularly Lasso regression, outperform more complex alternatives. The selected model achieves Spearman IC = 0.269, directional accuracy = 57.86%, and strong simulated trading performance with controlled risk.

CHAPTER IV. CONCLUSION

1. Summary of results

This study successfully developed a multi-source data-driven 4-hour bitcoin return prediction system. The system consists of three main components.

First, a three-layer ETL architecture was implemented to collect data from seven sources (OHLCV, market capitalization, network activity, mining, on-chain metrics, news, and sentiment index). The pipeline applies source-specific missing-data handling and performs news sentiment analysis using CryptoBERT. It supports incremental data collection, resuming from interruptions, and prevents future information leakage via an END_TIME mechanism.

Second, a multidimensional feature set of 31 variables was constructed across five main groups: price action (5 variables), microstructure (4 variables), technical indicators (2 variables), market structure (3 variables), on-chain health (8 variables), and sentiment (4 variables). All features are stationary, standardized using an ex-ante design to avoid future leakage, and do not exhibit severe multicollinearity.

Third, the Lasso regression model achieved strong performance on the test set (unseen data): Spearman IC = 0.269, directional accuracy = 57.86%, total PnL = +5.28 billion USD (+528% on the initial capital of USD 1 billion), maximum drawdown = -9%, and win rate = 57.86%. Compared with the Last baseline, the model improves by 1,078% in PnL (from -0.54 to +5.28 billion USD), 638% in Spearman IC (from -0.050 to 0.269), and reduces risk by 8.8× (MDD from -79% to -9%).

The model generalizes well on unseen data: the degradation from validation to test is mild (IC -10%, PnL -7.2%), while directional accuracy remains stable at 57.86% and maximum drawdown improves (-9% vs. -11%), indicating no concerning overfitting.

2. Research contributions

This study makes three primary contributions. Methodologically, it provides a comprehensive workflow for cryptocurrency price forecasting, covering data collection, modeling, and evaluation, and can be adapted to other financial time-series forecasting tasks.

Technically, it integrates five key feature groups (price action, market structure, on-chain health, on-chain valuation, and sentiment) into a unified framework that leverages both traditional data and alternative data (on-chain metrics and sentiment analysis). The use of ex-ante rolling statistics represents a best practice for preventing future leakage.

Empirically, the study demonstrates that regularized linear models (Lasso, Ridge, Huber) substantially outperform nonlinear and ensemble approaches in this crypto forecasting setting. This supports the view that in financial markets with weak signals

and high noise, simple but well-regularized models can be more effective than complex alternatives.

3. Limitations

Despite strong results, several limitations remain. First, the trading simulation does not include transaction fees ($\sim 0.1\%$ per trade) or slippage, which could materially reduce realized PnL. Second, the evaluation is limited to bitcoin on the 4-hour timeframe; performance may differ for Ethereum, altcoins, or other horizons. Third, the model is static—its weights are not updated with new data—creating potential model drift over time. Fourth, deep learning and reinforcement learning approaches were not explored, which may offer additional performance gains.

4. Future work

To improve the system, several directions are proposed: (1) implementing online learning to periodically update model weights; (2) extending to multi-asset (Ethereum, altcoins) and multi-timeframe settings (1H, 1D, 1W); (3) incorporating transaction costs into the simulation; (4) testing deep learning or reinforcement learning approaches; (5) building a monitoring system to detect model drift; and (6) applying dynamic position sizing and more advanced risk management.

5. Conclusion

This study shows that combining multi-source data with machine learning models is a viable approach for short-term cryptocurrency forecasting. The Lasso model achieves strong performance with positive PnL (+528%), low risk (-9% maximum drawdown), and robust generalization on unseen data. The ETL pipeline is designed to handle heterogeneous, multi-frequency data sources in a reliable manner. Although limitations remain, the positive results highlight the practical potential of this approach. With further enhancements (online learning, multi-asset extension, deep learning), the system may achieve even stronger performance.

REFERENCE

APIs and data sources

Binance API: <https://binance-docs.github.io/apidocs/spot/en/>

CoinGecko API: <https://www.coingecko.com/en/api/documentation>

Glassnode API: <https://docs.glassnode.com/api/>

CryptoQuant API: <https://cryptoquant.com/api>

CoinMarketCap API: <https://coinmarketcap.com/api/documentation/>

Alternative.me API: <https://alternative.me/crypto/fear-and-greed-index/api/>

Libraries and tools

Pandas: <https://pandas.pydata.org/docs/>

NumPy: <https://numpy.org/doc/>

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NLP and CryptoBERT

Hugging Face Transformers: <https://huggingface.co/transformers/>

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BERT Paper: <https://arxiv.org/abs/1810.04805>

APENDIX

A. Detailed list of raw variables and definitions

Group	Variable name	Definition	Unit
OHLCV (BTC)	BTC_open	BTC opening price in the 4H interval	USD
	BTC_high	BTC highest price in the 4H interval	USD
	BTC_low	BTC lowest price in the 4H interval	USD
	BTC_close	BTC closing price in the 4H interval	USD
	BTC_volume	BTC trading volume (base volume)	BTC
	BTC_quote_volume	Quote trading volume (USDT)	USDT
	BTC_trades	Number of executed trades in the interval	count
OHLCV (ETH)	ETH_open/high/low/close	ETH OHLC in the 4H interval	USD
	ETH_volume/quote_volume/trades	ETH volume and trade count	–
Market cap	BTC_market_cap	BTC market capitalization	USD
	ETH_market_cap	ETH market capitalization	USD

	USDT_market_cap	USDT market cap (approx. total supply)	USD
	USDC_market_cap	USDC market cap (approx. total supply)	USD
Network activity	BTC_active_addresses	Number of active BTC addresses in the interval	addresses
	BTC_tx_count	Number of BTC on-chain transactions	transactions
	ETH_active_addresses	Number of active ETH addresses	addresses
	ETH_tx_count	Number of ETH on-chain transactions	transactions
Mining	mining_difficulty	BTC mining difficulty	difficulty
	hash_rate_ths	Network hash rate	TH/s
	miner_revenue_usd	Miner revenue (block reward + fees)	USD
On-chain metrics	BTC_onchain_total_supply	Circulating BTC supply	BTC
	BTC_onchain_mrvr_ratio	Market value / realized value	ratio
	BTC_onchain_exchange_inflow_usd	BTC inflow to exchanges (in USD)	USD
	BTC_onchain_exchange_outflow_usd	BTC outflow from exchanges (in USD)	USD

	BTC_onchain_exchange_supply_native	BTC balance held on exchange wallets	BTC
News sentiment	news_article_count	Number of news articles in the 4H interval	count
	news_combined_sent_mean	Mean combined sentiment	-1 to +1
	news_bull/bear_ratio	Bullish-to-bearish ratio	0-1
	news_p_bull/bear/neu_mean	Mean CryptoBERT probabilities	0-1
	news_head_exag_ratio	“Clickbait” ratio (title > content)	ratio
Market sentiment	sentiment_index_value	Fear & Greed Index (0-100)	0-100
	sentiment_index_classification	Sentiment category	5 levels

B. List of 31 predictors and the target

No.	Feature	Group	Economic meaning	Unit / range
1	target	target	Future 4-hour BTC return	log return (~0)
2	BTC_log_return	OHLCV	Current BTC return	log return
3	ETH_log_return	OHLCV	Current ETH return	log return

4	BTC_vol_log_return	volume	BTC volume momentum	log change
5	log_rvol	volume	Relative volume (log)	relative z-score
6	vol_gk	volatility	Garman–Klass realized volatility	positive
7	vol_gk_z	volatility	Regime-normalized volatility	z-score
8	vol_signed	volatility	Signed volatility (buy/sell pressure)	$\pm$
9	clv	microstructure	Close location within the candle	$[-1, +1]$
10	wick_imbalance	microstructure	Upper/lower wick imbalance	$[-1, +1]$
11	bb_pct_b	technical	Price position within Bollinger bands	$[0, 1]$ (may overflow)
12	bb_width	technical	Bollinger band width	positive
13	btc_eth_corr	inter-market	Rolling BTC–ETH correlation	$[-1, +1]$
14	log_rel_ats	order flow	Normalized average trade size	log ratio
15	stable_supply_change	market cap	Stablecoin flow proxy	log change
16	ssr_z_score	market cap	BTC cap / stable cap (standardized)	z-score
17	usdt_dom_change	market cap	Change in USDT dominance	difference

18	addr_log_growth	network	Active address momentum	log change
19	intensity_change	network	Change in transaction intensity	difference
20	network_valuation_z	network	Standardized price-to-usage valuation	z-score
21	hash_ribbon_spread	mining	Hash ribbon (capitulation/recovery)	ratio
22	log_puell	mining	Puell multiple (miner selling pressure)	log ratio
23	onchain_mrvv_z	on-chain	Standardized MVRV ratio	z-score
24	onchain_netflow_z	on-chain	Standardized exchange netflow	z-score
25	reserve_ratio_change	on-chain	Change in exchange reserve ratio	difference
26	news_article_count	news	Article count	count
27	news_impact_score	news	News impact (sentiment $\times$ attention)	weighted score
28	news_sentiment_decay_slow	news	Smoothed sentiment decay (ewm span=6)	smoothed score
29	news_sentiment_shock	news	Abrupt sentiment change	difference
30	fng_change	sentiment	Fear & Greed momentum	difference
31	fng_divergence	sentiment	Divergence vs. MA7	difference

C. Mapping from raw data groups to engineered features

Raw feature group	# raw features	# engineered features	Selected engineered features
OHLCV (price/return)	14 (BTC+ETH)	3	BTC_log_return, ETH_log_return, target
Volume	6 (BTC+ETH)	2	BTC_vol_log_return, log_rvol
Volatility	0 (derived from OHLC)	3	vol_gk, vol_gk_z, vol_signed
Microstructure	0 (derived from OHLC)	2	clv, wick_imbalance
Technical indicators	0 (derived from OHLC)	2	bb_pct_b, bb_width
Inter-market correlation	0 (derived from returns)	1	btc_eth_corr
Order flow	4 (quote vol, trades)	1	log_rel_at
Market cap	4	3	stable_supply_change, ssr_z_score, usdt_dom_change
Network activity	4	3	addr_log_growth, intensity_change, network_valuation_z
Mining metrics	3	2	hash_ribbon_spread, log_puell

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On-chain metrics	8	3	onchain_mvrz_z, onchain_netflow_z, reserve_ratio_change
News sentiment	23	4	news_article_count, news_impact_score, news_sentiment_decay_slow, news_sentiment_shock
Market sentiment	2	2	fng_change, fng_divergence
Total	68	31	32 features (31 predictors + target)